

Industrial University of Industry (IUH)



Chapter 4. Digital communications

4.1

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3/4/2025

Overview

Communication is the transfer of information from one place to another.

This should be done

- as **efficiently** as possible
- with as much **fidelity/reliability** as possible
- as **securely** as possible

Communication System: Components/subsystems act together to accomplish information transfer/exchange.

Introduction

- **Source: analog or digital**
- **Transmitter: transducer, amplifier, modulator, oscillator, power amp., antenna**
- **Channel: e.g. cable, optical fibre, free space**
- **Receiver: antenna, amplifier, demodulator, oscillator, power amplifier, transducer**
- **Recipient: e.g. person, (loud) speaker, computer**

➤ Types of information:

Voice, data, video, music, email etc.

➤ Types of communication systems:

Public Switched Telephone Network
(voice, fax, modem)

Satellite systems

Radio, TV broadcasting

Cellular phones

Computer networks (LANs, WANs, WLANs)

Brief Chronology of Communication Systems

- **Two developments in the field of electricity**
 - ✓ First, in 1800, the Italian physicist Alessandro Volta invented the battery
 - ✓ Second, in 1820, the Danish physicist Hans Christian Oersted demonstrated the connection between electricity and magnetism
- **In May 24, 1844 Samuel Morse transmitted the first public telegraph message**



Brief Chronology of Communication Systems

- 1874 Thomas Edison invented quadruplex telegraphy
- 1876 Alexander Graham Bell invented the telephone
- 1906 First wireless communication of human speech
- 1936 First television broadcast made in London, England
- 1938-1945 *World War II* Radar and microwave systems
- 1941 Z3 computer developed by German engineer Konrad Zuse
- 1948-1950 *Information Theory and coding*. C. E. Shannon
- 1957 *Satellite* communications begins with Sputnik I. Russia
- 1962-1966 *High Speed digital communication*
- 1972 Motorola develops *cellular telephone*.
- 1994 High Definition TV standards agreed on in the U.S.
- 1999 Intel introduces the Pentium III processor, taking personal computing to new highs.
- 2050??? Tell me where you think we will be in the year 2050

Types of communication

verbal, nonverbal...

1 Analogue communication

modulation: amplitude modulation (am), frequency modulation (fm), phase modulation (pm)

2 Digital communication

modulation: Frequency Shift Keying (FSK), Binary Phase Shift Keying (BPSK), Quadrature Phase Shift Keying (QPSK)

WHY DIGITAL?

- **Digital techniques need to distinguish between discrete symbols allowing regeneration versus amplification**
- **Good processing techniques are available for digital signals, such as medium.**
 - Data compression (or source coding)
 - Error Correction (or channel coding)(A/D conversion)
 - Equalization
 - Security
- **Easy to mix signals and data using digital techniques**

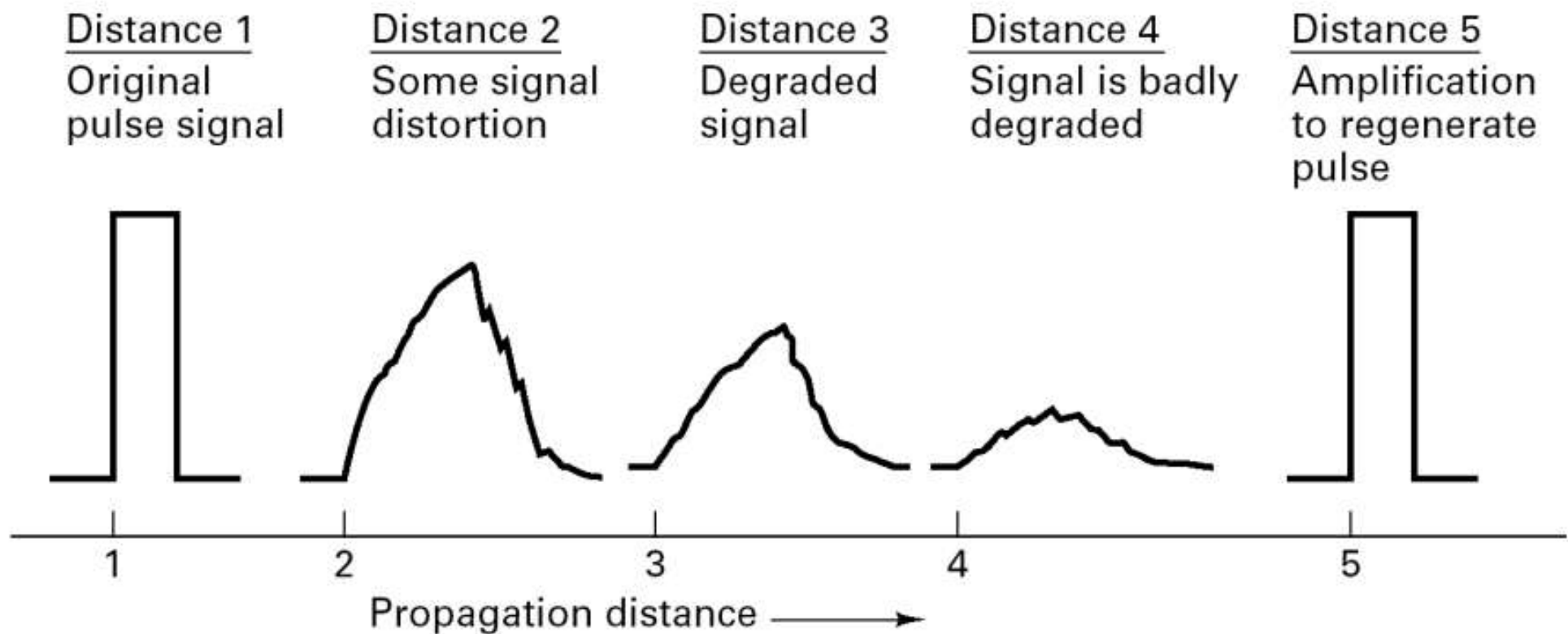


Figure 1.1 Pulse degradation and regeneration.

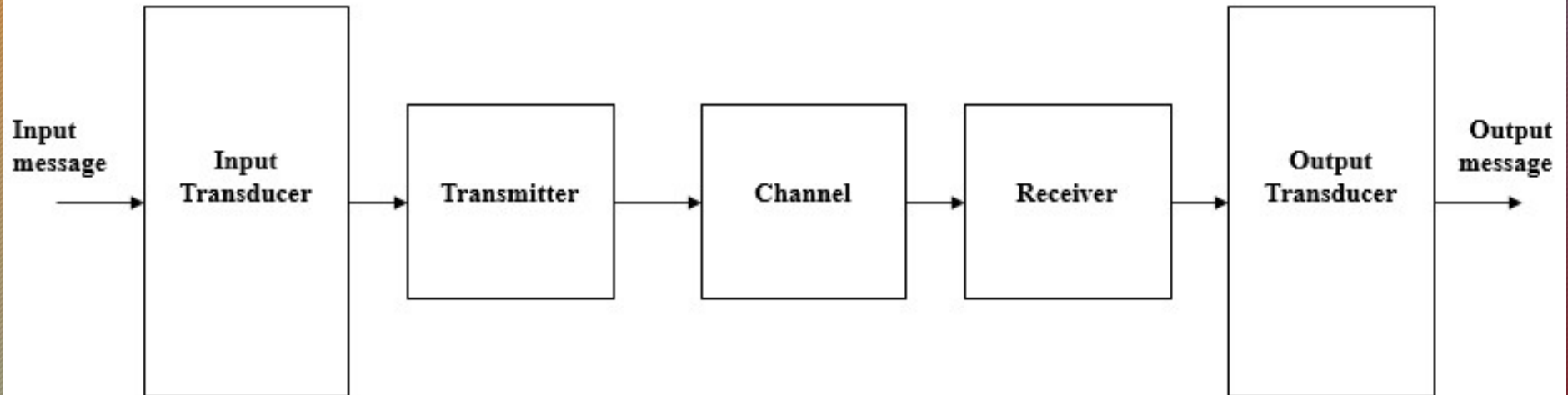
Telecommunication

- Telegraph
- Fixed line telephone
- Cable
- Wired networks
- Internet
- Fiber communications
- Communication bus inside computers to communicate between CPU and memory

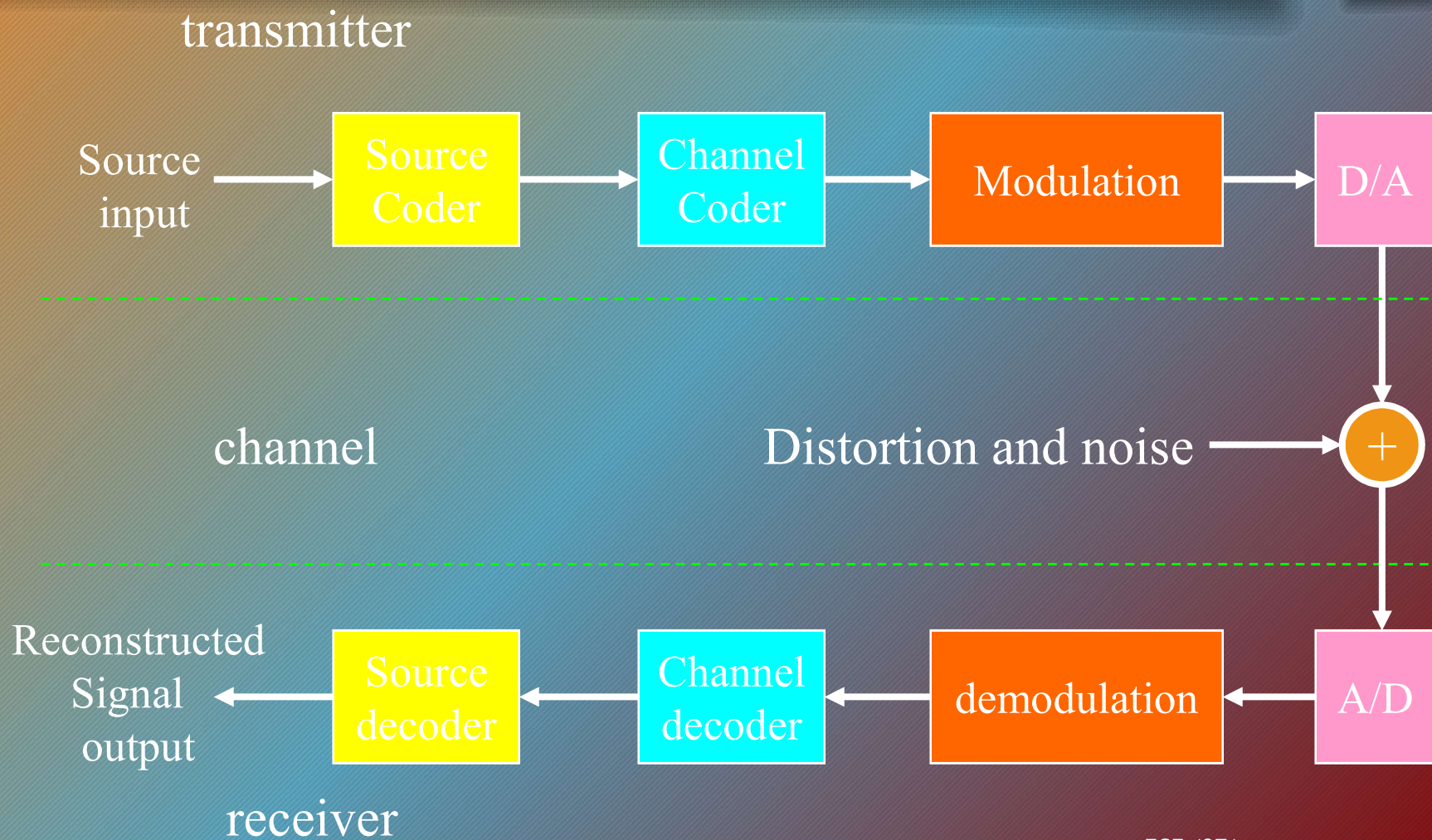
Wireless Communications

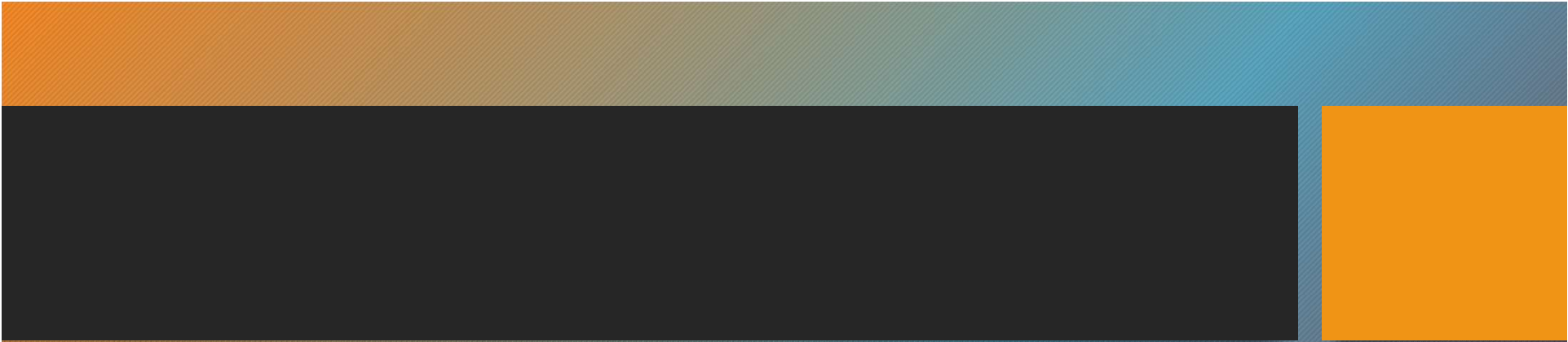
- Satellite
- TV
- Cordless phone
- Cellular phone
- Wireless LAN, WIFI
- Wireless MAN, WIMAX
- Bluetooth
- Ultra Wide Band
- Wireless Laser
- Microwave
- GPS
- Ad hoc/Sensor Networks

Elements of a Communication System



Communication System Components



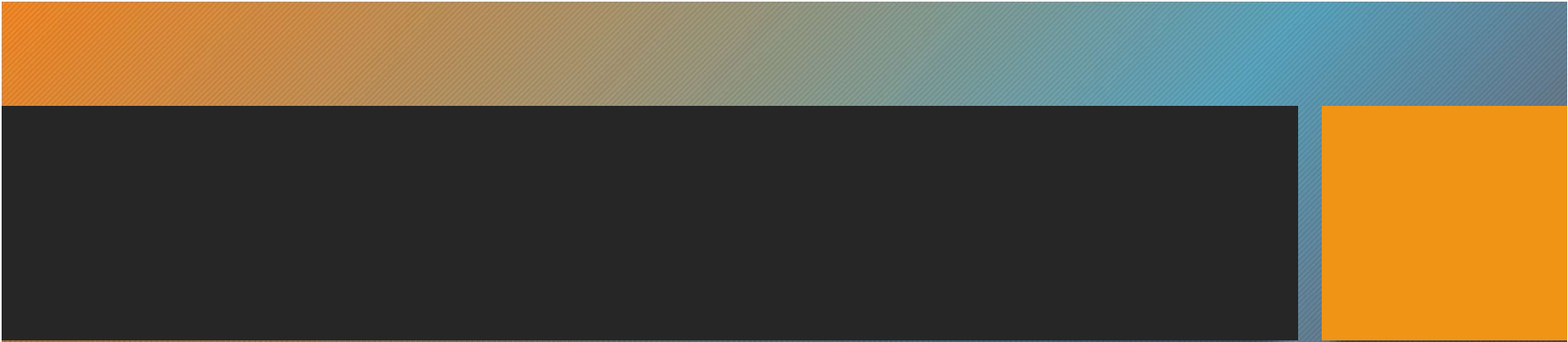


Input Transducer: The message produced by a source must be converted by a transducer to a form suitable for the particular type of communication system.

Example: In electrical communications, speech waves are converted by a microphone to voltage variation.

Transmitter: The transmitter processes the input signal to produce a signal suits to the characteristics of the transmission channel.

Signal processing for transmission almost always involves modulation and may also include coding. In addition to modulation, other functions performed by the transmitter are amplification, filtering and coupling the modulated signal to the channel.



Channel: The channel can have different forms: The atmosphere (or free space), coaxial cable, fiber optic, waveguide, etc.

The signal undergoes some amount of degradation from noise, interference and distortion

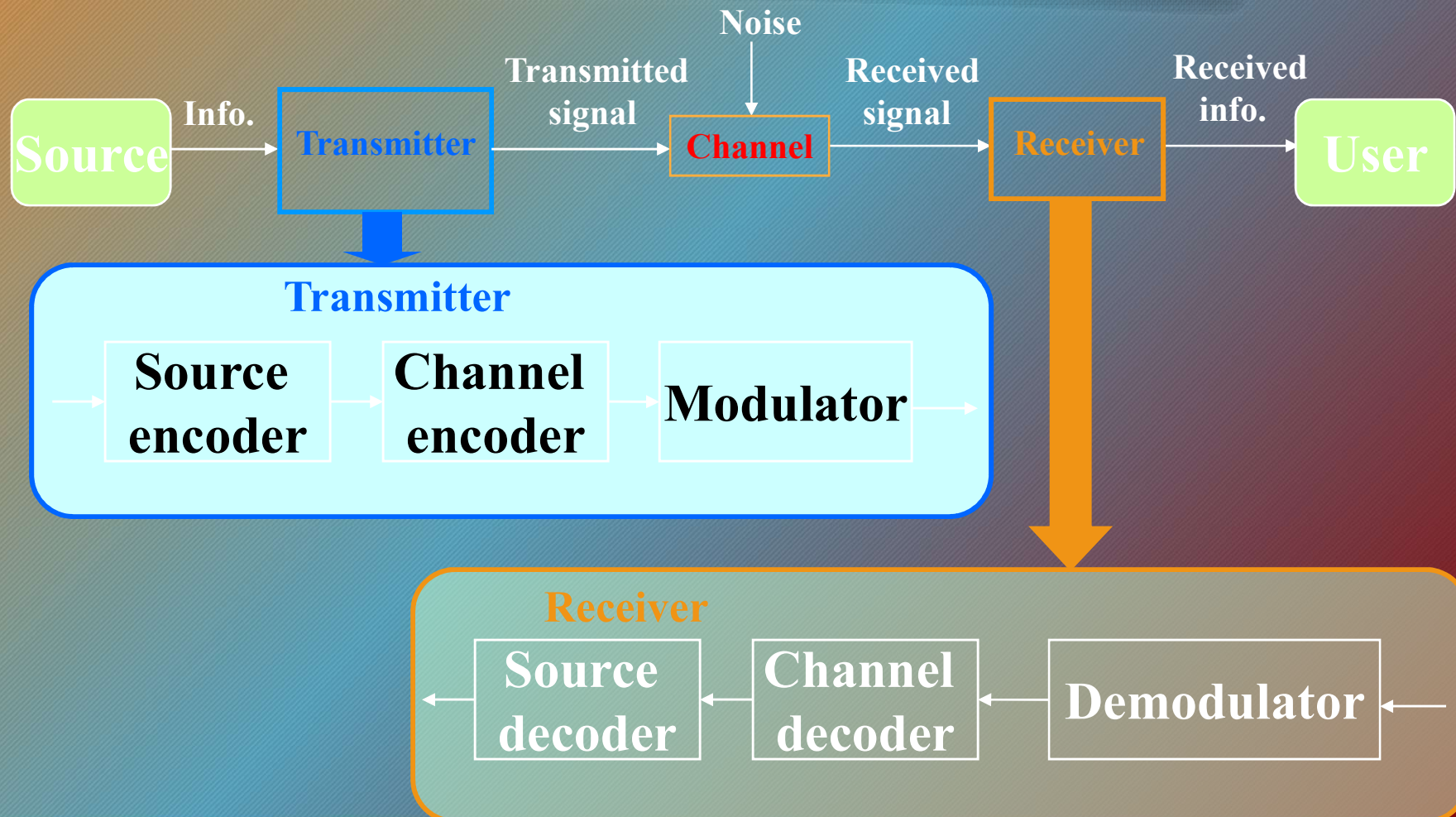
Receiver: The receiver's function is to extract the desired signal from the received signal at the channel output and to convert it to a form suitable for the output transducer.

Other functions performed by the receiver: amplification (the received signal may be extremely weak), demodulation and filtering.

Output Transducer: Converts the electric signal at its input into the form desired by the system user.

Example: Loudspeaker, personal computer (PC), tape recorders.

Elements of a digital communication systems



Source Encoder And Decoder

Source Encoder

- In digital communication we convert the signal from source into digital signal.
- ***Source Encoding or Data Compression:*** the process of efficiently converting the output of either analog or digital source into a sequence of binary digits is known as source encoding.

Source Decoder

- At the end, if an analog signal is desired then source decoder tries to decode the sequence from the knowledge of the encoding algorithm.

Channel Encoder And Decoder

Channel Encoder:

- The information sequence is passed through the channel encoder. The purpose of the channel encoder is to introduce, in controlled manner, some redundancy in the binary information sequence that can be used at the receiver to overcome the effects of noise and interference encountered in the transmission on the signal through the channel.

Channel Decoder:

- Channel decoder attempts to reconstruct the original information sequence from the knowledge of the code used by the channel encoder and the redundancy contained in the received data

Digital Modulator And Demodulator

Digital Modulator:

- The binary sequence is passed to digital modulator which in turns convert the sequence into electric signals so that we can transmit them on channel. The digital modulator maps the binary sequences into signal wave forms.

Digital Demodulator:

- The digital demodulator processes the channel corrupted transmitted waveform and reduces the waveform to the sequence of numbers that represents estimates of the transmitted data symbols.

Why Digital Communications?

- Easy to regenerate the distorted signal
- Regenerative repeaters along the transmission path can detect a digital signal and retransmit a new, clean (noise free) signal
- These repeaters prevent accumulation of noise along the path

This is not possible with analog communication systems

- Two-state signal representation

The input to a digital system is in the form of a sequence of bits (binary or M-ary)

- Immunity to distortion and interference
- Digital communication is rugged in the sense that it is more immune to channel noise and distortion

- Hardware is more flexible
- Digital hardware implementation is flexible and permits the use of microprocessors, mini-processors, digital switching and VLSI

Shorter design and production cycle

- Low cost

The use of LSI and VLSI in the design of components and systems have resulted in lower cost

- Easier and more efficient to multiplex several digital signals

Digital multiplexing techniques – Time & Code Division Multiple Access - are easier to implement than analog techniques such as Frequency Division Multiple Access

- Can combine different signal types – data, voice, text, etc.
 - Data communication in computers is digital in nature whereas voice communication between people is analog in nature
 - **Using digital techniques, it is possible to combine both format for transmission through a common medium**
 - **Encryption and privacy techniques are easier to implement**
 - Better overall performance
 - Digital communication is inherently more efficient than analog in realizing the exchange of SNR for bandwidth.
- Digital signals can be coded to yield extremely low rates and high fidelity as well as privacy.

Disadvantages:

- **Requires reliable “synchronization”**
- **Requires A/D conversions at high rate**
- **Requires larger bandwidth**
- **Nongraceful degradation**
- **Performance Criteria**
- **Probability of error or Bit Error Rate**

ADC/DAC

- Analog-to-Digital Conversion (ADC) and Digital-to-Analog Conversion (DAC) are the processes that allow digital computers to interact with these everyday signals.
- Digital information is different from its continuous counterpart in two important respects: it is *sampled*, and it is *quantized*

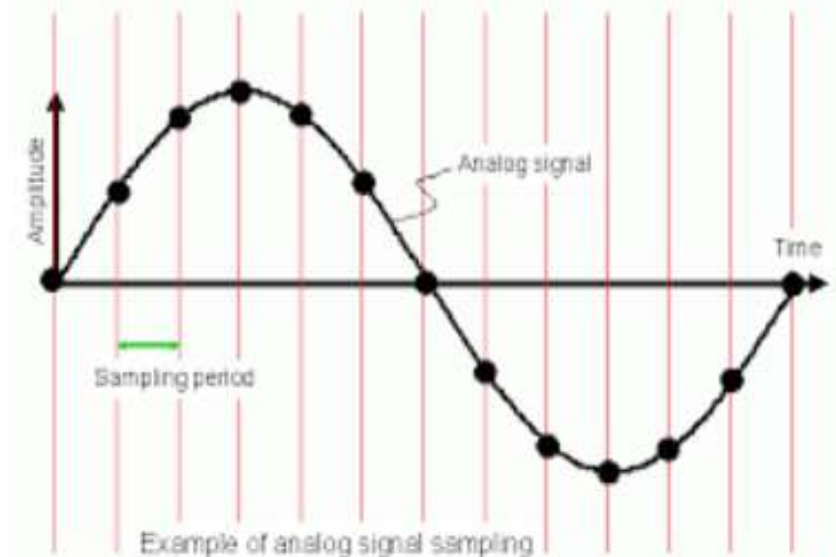


Sampling Process

Sampling is converting a *continuous time* signal into a *discrete time* signal.

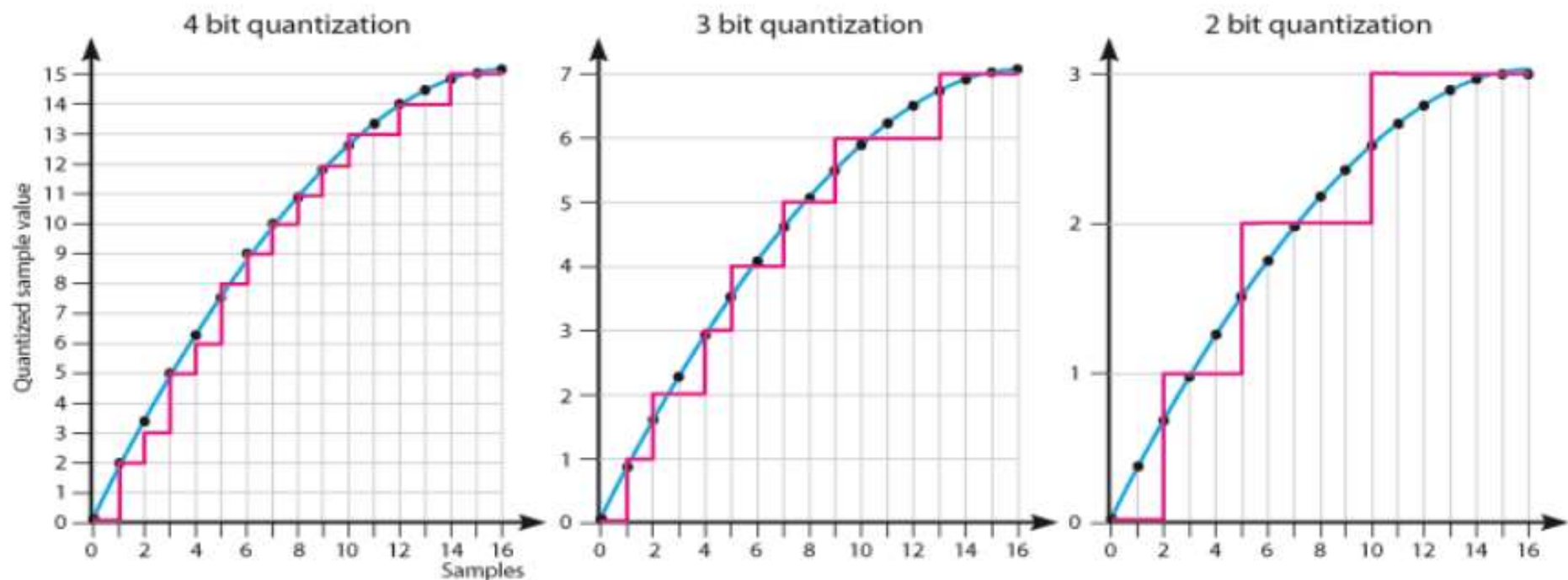
There three types of sampling

- Impulse (ideal) sampling
- Natural Sampling
- Sample and Hold operation



Quantization

Quantization is a non linear transformation which maps elements from a continuous set to a finite set.



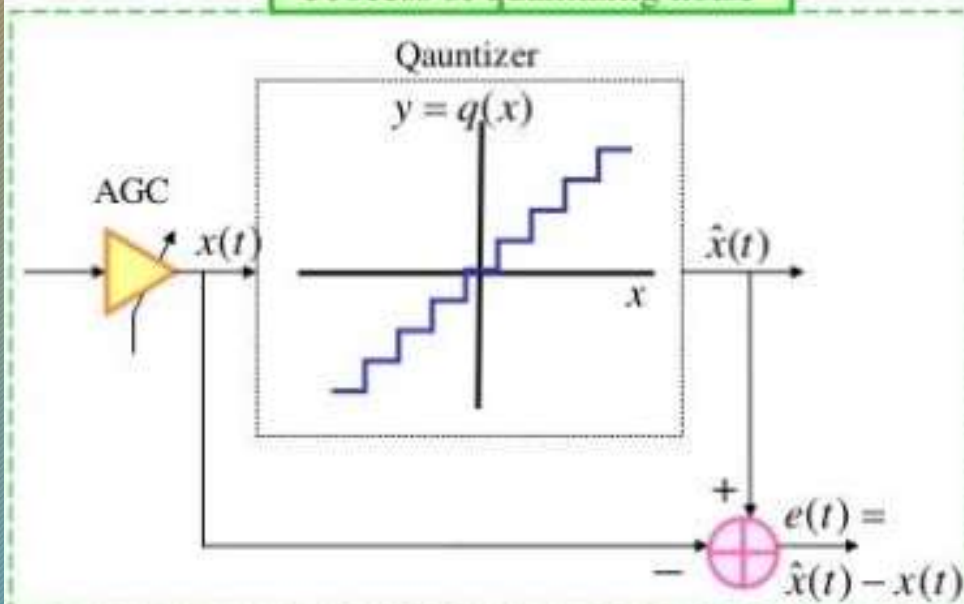
Quantization Noise

Quantization error

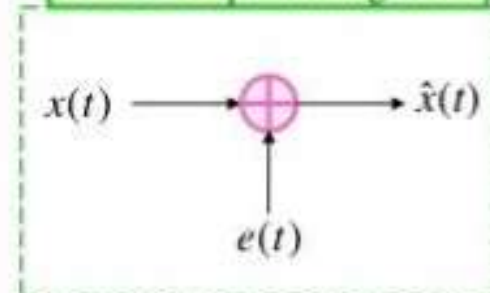
- Quantizing error: The difference between the input and output of a quantizer

$$\Rightarrow e(t) = \hat{x}(t) - x(t)$$

Process of quantizing noise



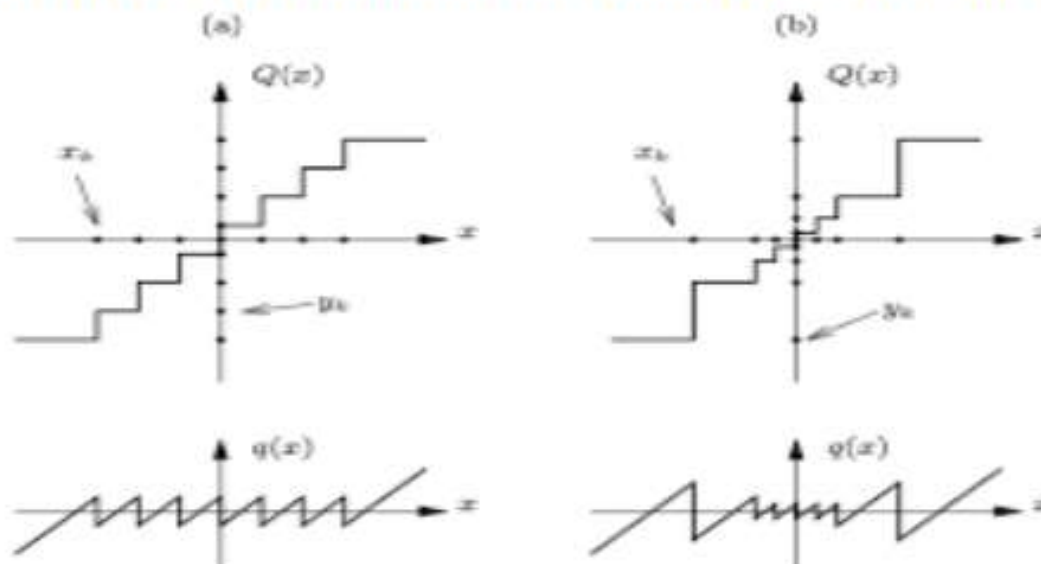
Model of quantizing noise



Uniform & Non-Uniform Quantization

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Non uniform Quantization Used to reduce quantization error and increase the dynamic range when input signal is not uniformly distributed over its allowed range of values.



Uniform quantization

Non-Uniform
quantization

Condition for Optimal Quantizer

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Find the two sets $\{v_k\}_{k=1}^L$ and $\{J_k\}_{k=1}^L$, that minimize the average distortion

$$D = \sum_{k=1}^L \int_{m \in J_k} d(m, v_k) f_M(m) dm$$

where $f_M(m)$ is the pdf, which is known.

The mean - square distortion is used commonly

$$d(m, v_k) = (m - v_k)^2$$

The optimization is a nonlinear problem which may not have closed form solution. However the quantizer consists of two components : an encoder characterized by J_k , and a decoder characterized by v_k

Encoding

PCM encoding example

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Level	Code word
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

Levels are encoded using this table

Table: Quantization levels with belonging code words

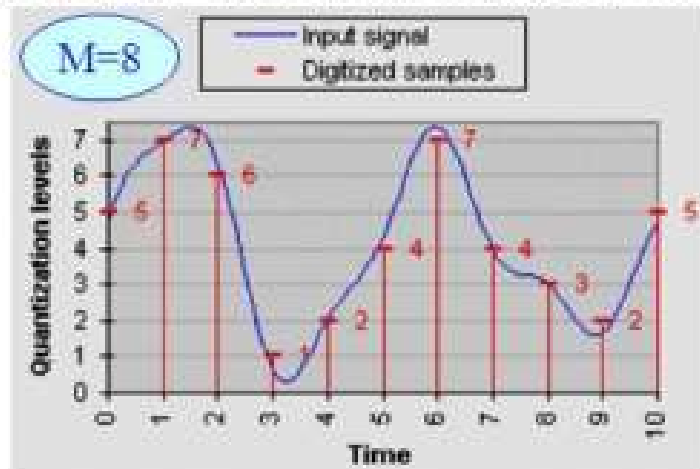


Chart 1. Quantization and digitalization of a signal.

Signal is quantized in 11 time points & 8 quantization segments.

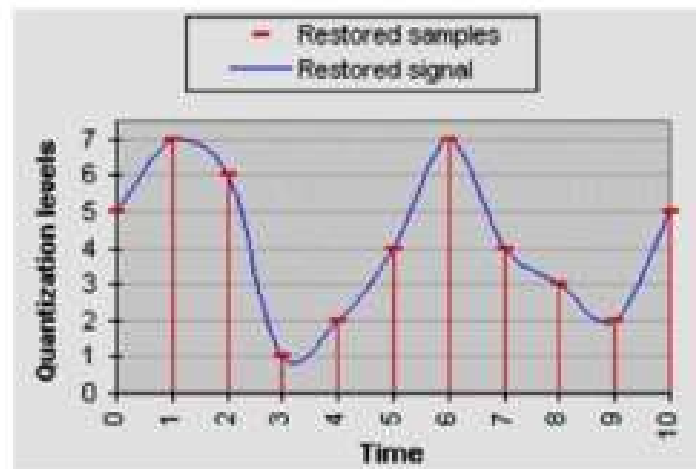


Chart 2. Process of restoring a signal.

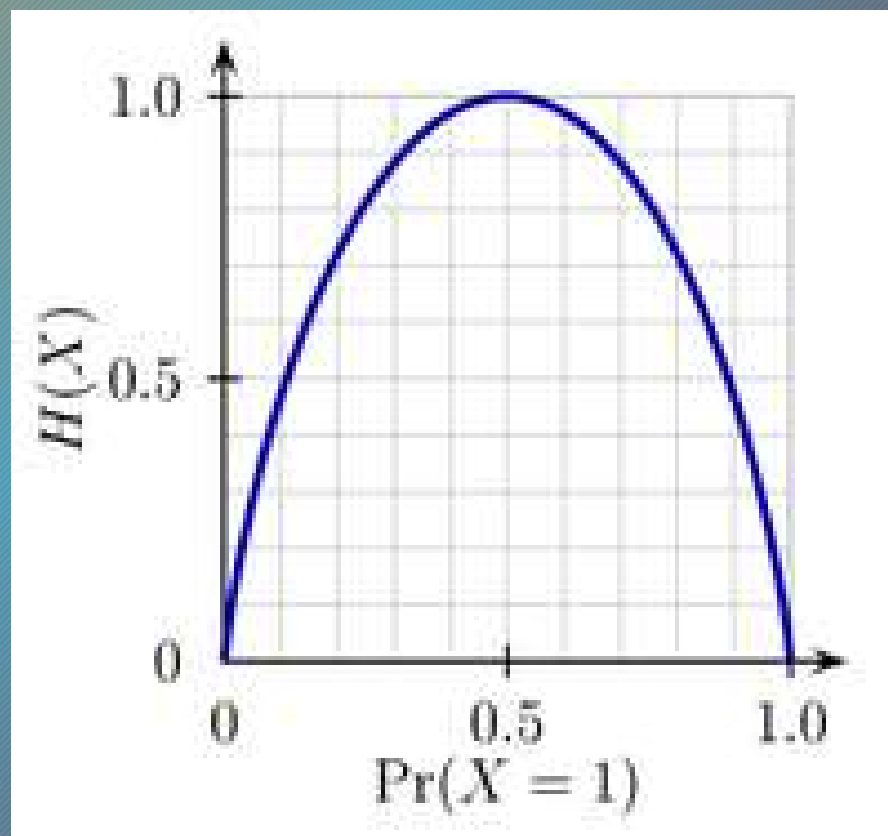
PCM encoded signal in binary form:

101 111 110 001 010 100 111 100 011 010 101

Total of 33 bits were used to encode a signal

Source Coder

- Examples
 - Digital camera: encoder;
TV/computer: decoder
 - Camcorder
 - Phone
 - Read the book
- Theorem
 - How much information is measured by **Entropy**
 - More randomness, high entropy and more information



$$H(x) = \sum_{i=1}^n p(i) \log_2 \left(\frac{1}{p(i)} \right) = - \sum_{i=1}^n p(i) \log_2 p(i)$$

Power, Channel, Noise

- Transmit power
 - Constrained by device, battery, health issue, etc.
- Channel responses to different frequency and different time
 - Satellite: almost flat over frequency, change slightly over time
 - Cable or line: response very different over frequency, change slightly over time.
 - Fiber: perfect
 - Wireless: worst. Multipath reflection causes fluctuation in frequency response. Doppler shift causes fluctuation over time
- Noise and interference
 - AWGN: Additive White Gaussian noise
 - Interferences: power line, microwave, other users (CDMA phone)

Shannon Capacity

- Shannon Theory

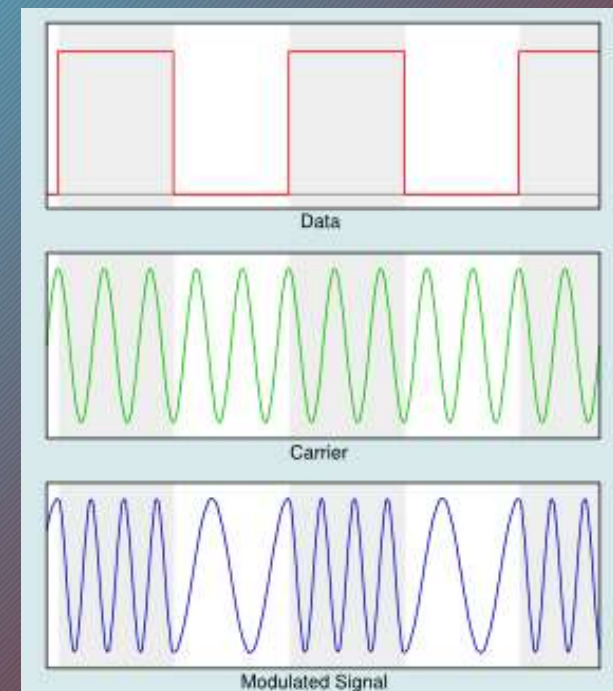
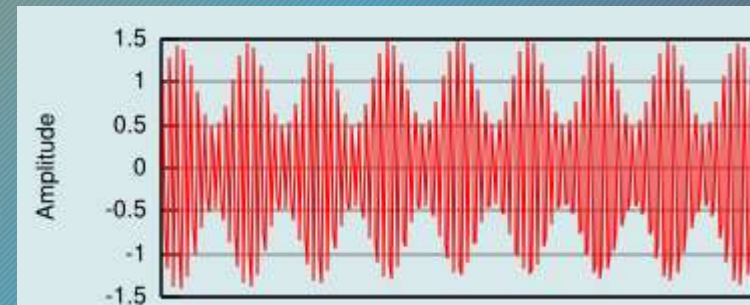
- It establishes that given a noisy channel with information capacity C and information transmitted at a rate R , then if $R < C$, there exists a coding technique which allows the probability of error at the receiver to be made arbitrarily small. This means that theoretically, it is possible to transmit information without error up to a limit, C .
- The converse is also important. If $R > C$, the probability of error at the receiver increases without bound as the rate is increased. So no useful information can be transmitted beyond the channel capacity. The theorem does not address the rare situation in which rate and capacity are equal.

- Shannon Capacity

$$C = B \log_2(1 + SNR) \quad \text{bit / s}$$

Modulation

- Process of varying a carrier signal in order to use that signal to convey information
 - Carrier signal can transmit far away, but information cannot
 - Modem: amplitude, phase, and frequency
 - Analog: AM, amplitude, FM, frequency, Vestigial sideband modulation, TV
 - Digital: mapping digital information to different constellation: Frequency-shift key (FSK)



Channel Coding

- Purpose
 - Deliberately add redundancy to the transmitted information, so that if the error occurs, the receiver can either detect or correct it.
- Source-channel separation theorem
 - If the delay is not an issue, the source coder and channel coder can be designed separately, i.e. the source coder tries to pack the information as hard as possible and the channel coder tries to protect the packet information.
- Popular coder
 - Linear block code
 - Cyclic codes (CRC)
 - Convolutional code (Viterbi, Qualcomm)
 - LDPC codes, Turbo code, 0.1 dB to Channel Capacity

Quality of a Link (service, QoS)

- Mean Square Error

$$MSE = \frac{1}{N} \sum_{i=1}^N |\hat{X}_i - X_i|^2$$

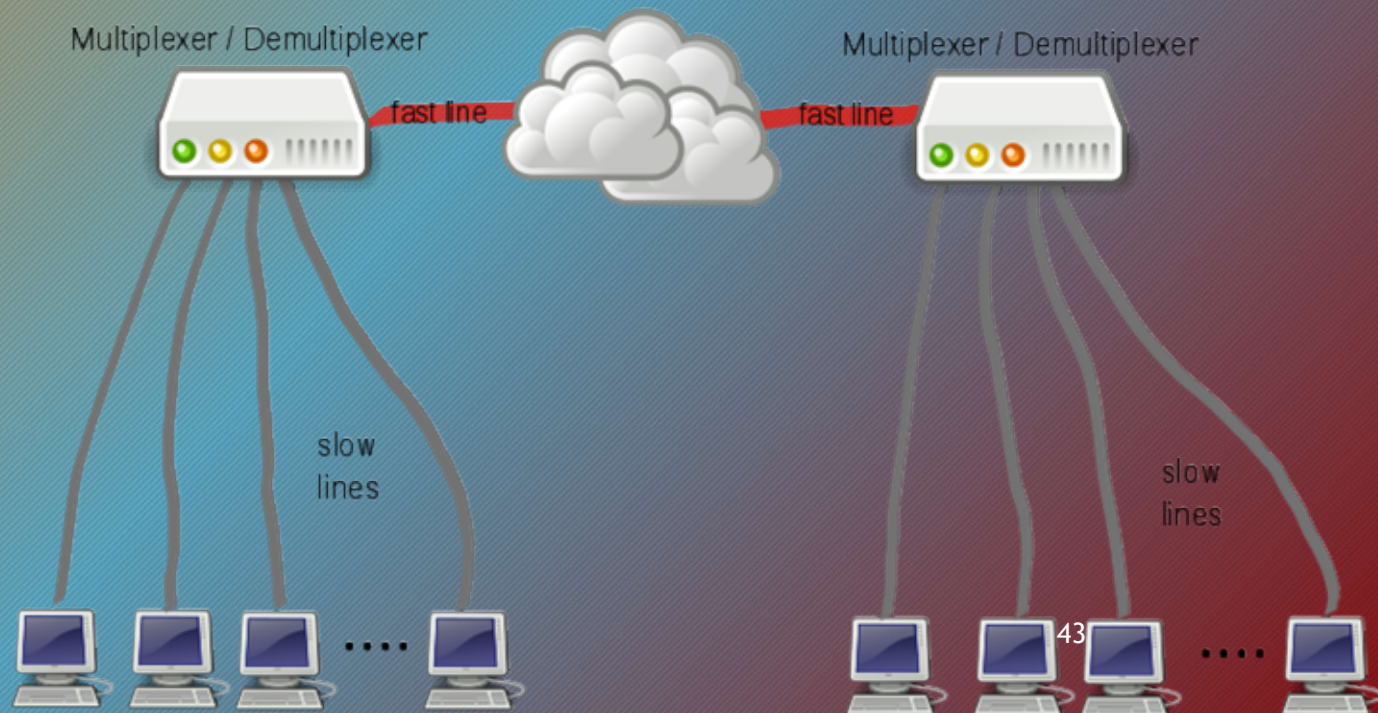
- Signal to noise ratio (SNR)

$$\Gamma = \frac{P_{rec}}{\sigma^2} = \frac{P_{tx} G}{\sigma^2}$$

- Bit error rate
 - Frame error rate
 - Packet drop rate
 - Peak SNR (PSNR)
 - SINR/SNIR: signal to noise plus interference ratio
- Human factor

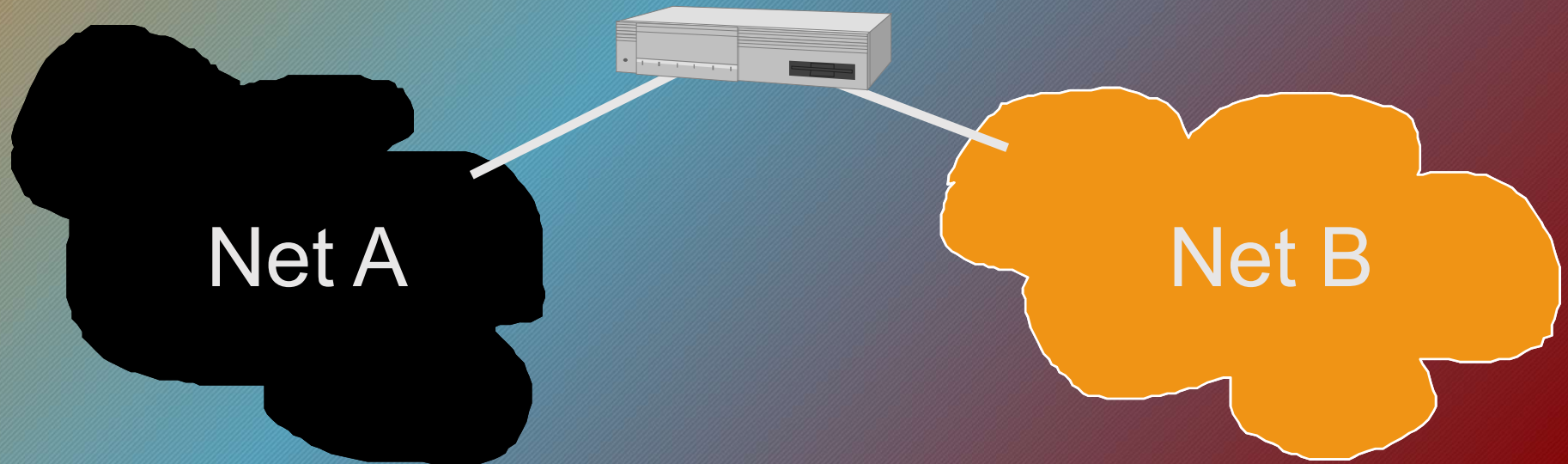
Multiplexing

- Space-division multiplexing
- Frequency-division multiplexing
- Time-division multiplexing
- Code-division multiplexing



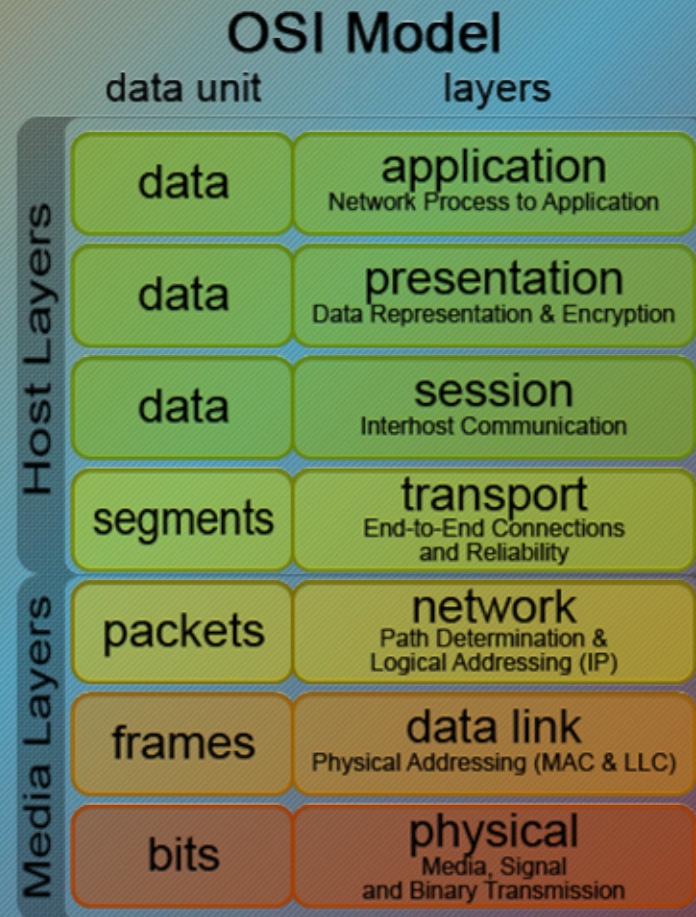
Communication Networks

- Connection of 2 or more distinct (possibly dissimilar) networks.
- Requires some kind of network device to facilitate the connection.
- Internet



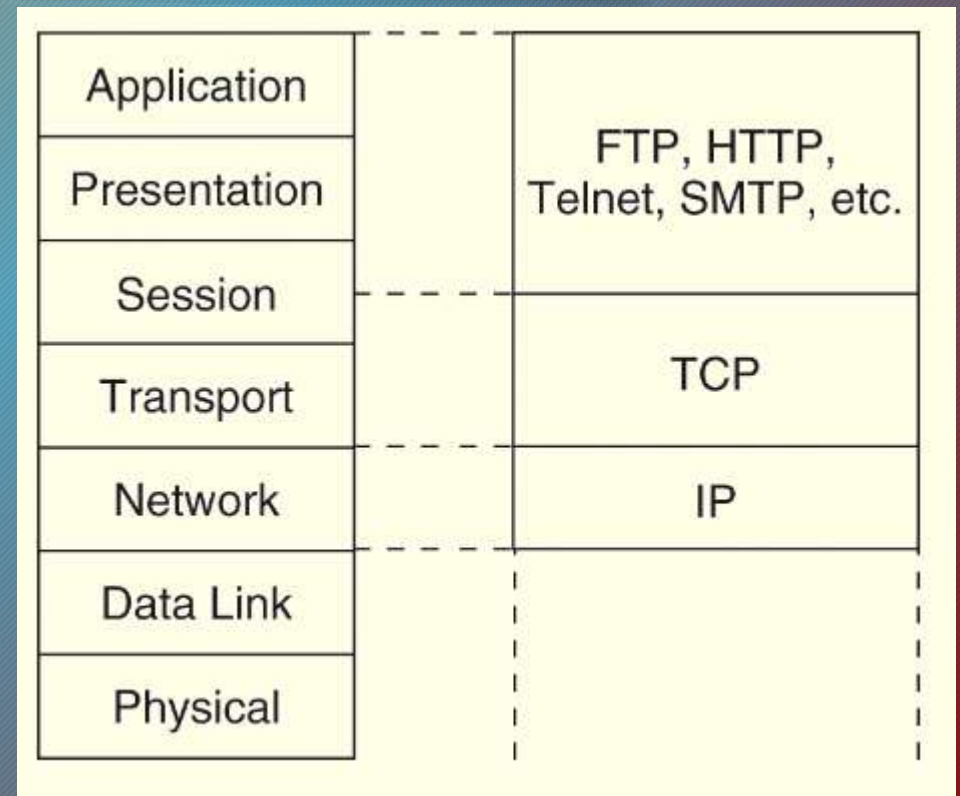
OSI Model

Open Systems Interconnections; Course offered next semester



TCP/IP Architecture

- TCP/IP is the de facto global data communications standard.
- It has a lean 3-layer protocol stack that can be mapped to five of the seven in the OSI model.
- TCP/IP can be used with any type of network, even different types of networks within a single session.

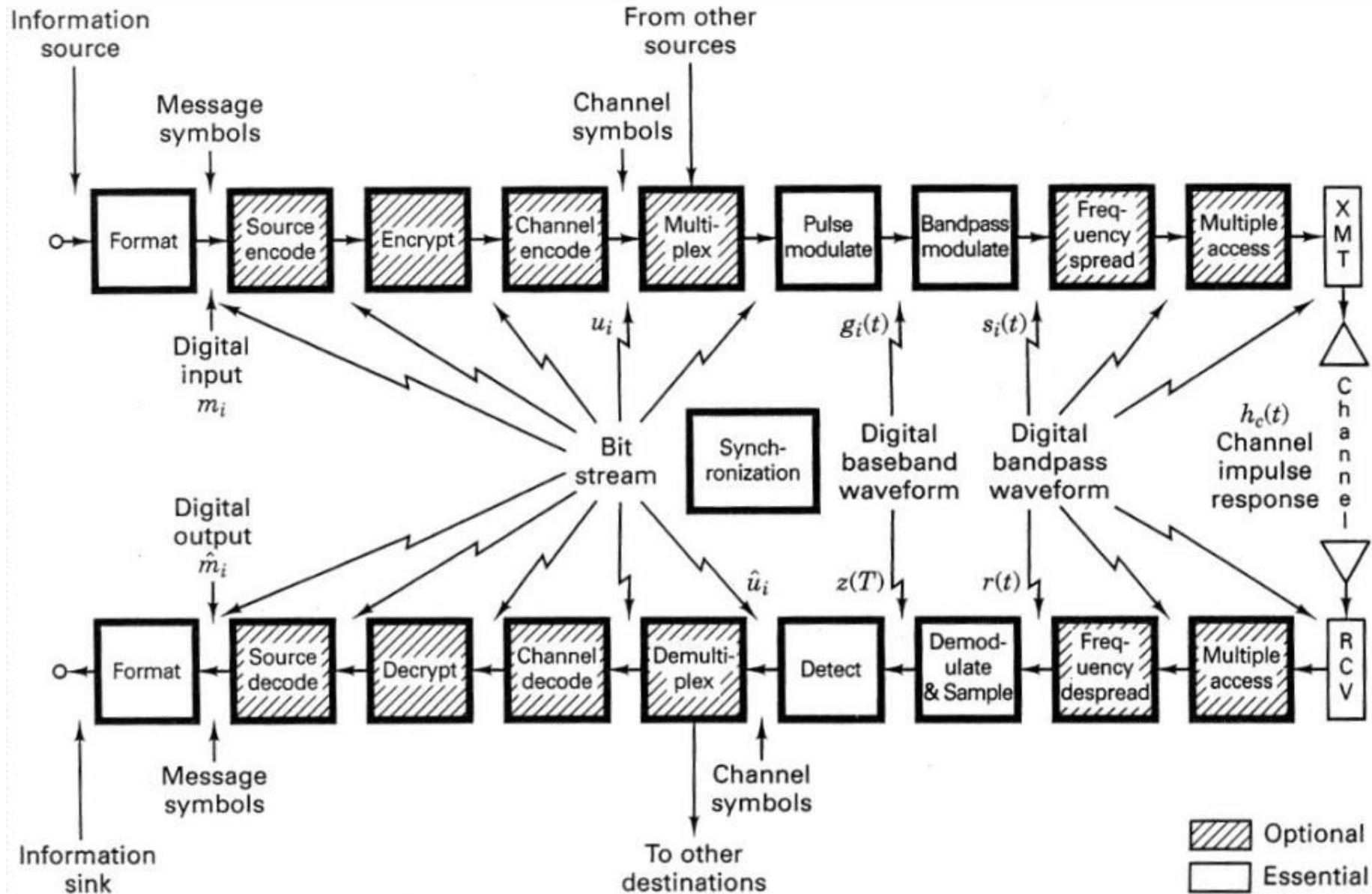


Outline

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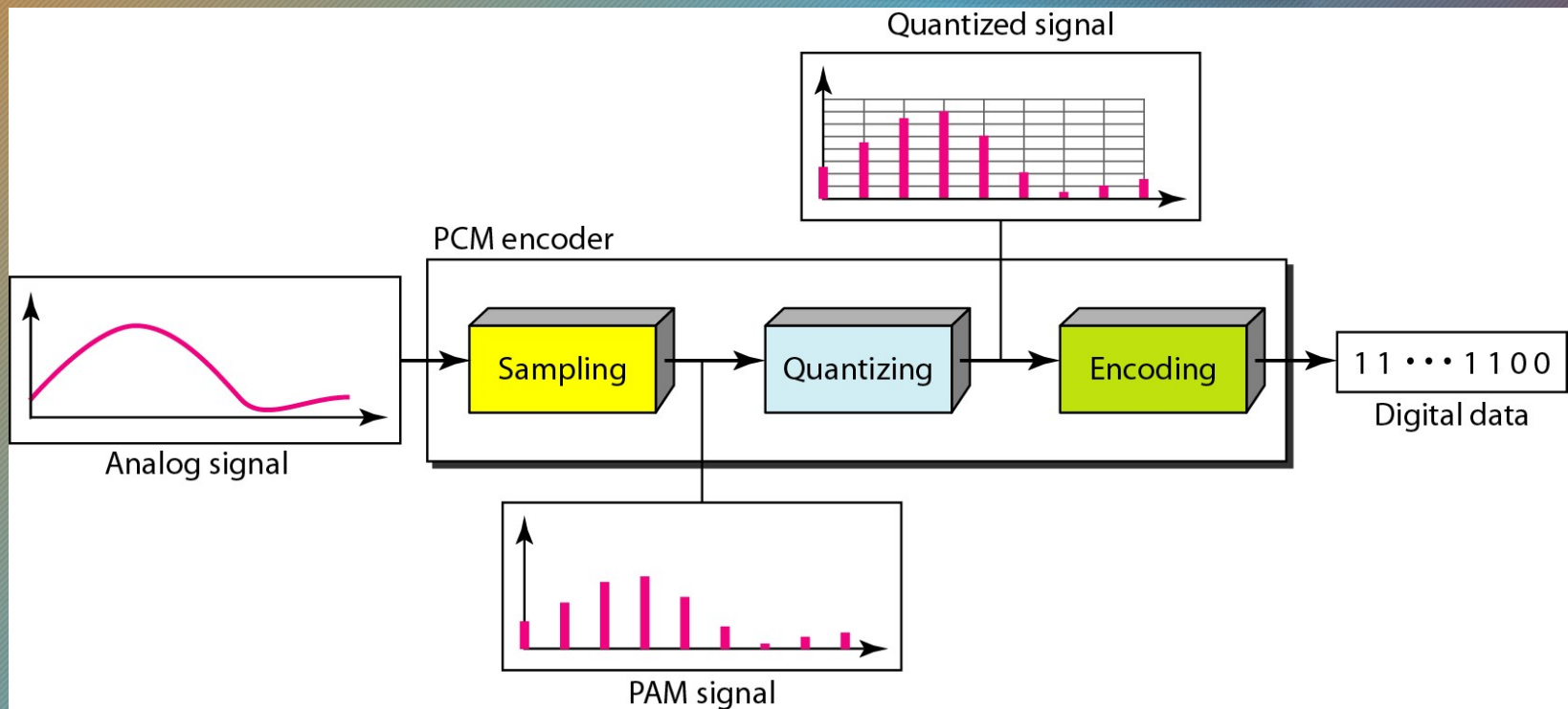
- I. Overview of digital communication network structure
- II. PCM/PDM
- III. Digital Transmission Modes
- IV. TDM for E1/T1 systems

Digital communications system block diagram



II. Pulse Code Modulation (PCM)

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Pulse-Code Modulation

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Pulse-Code Modulation (PCM) is a digital transmission system with an **analog-to digital converter (ADC)** at the input and a **digital-to-analog converter (DAC)** at the output.

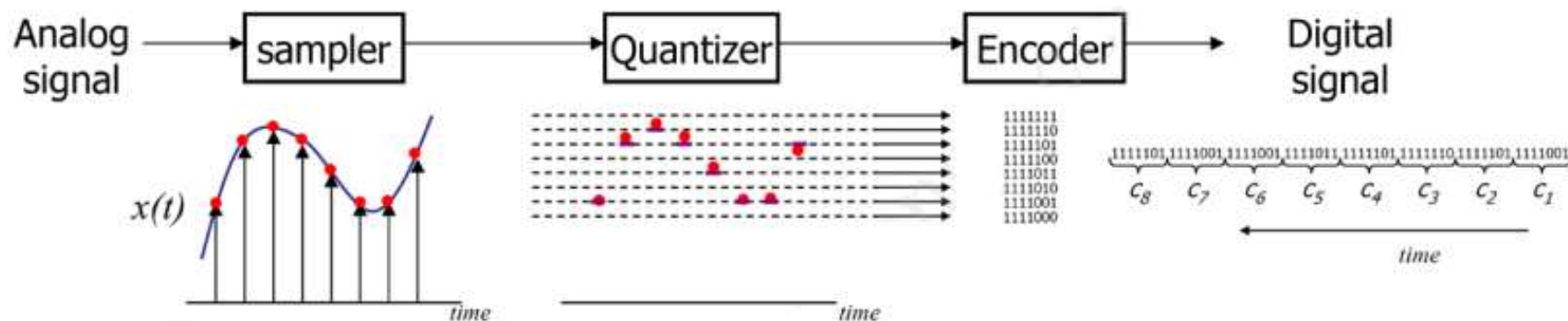
Pulse Code Modulation

- **Pulse Code Modulation (PCM) is a special form of A/D conversion.**
- **It consists of sampling, quantizing, and encoding steps.**
 - 1.Used for long time in telephone systems**
 - 2. Errors can be corrected during long haul transmission**
 - 3. Can use time division multiplexing**
 - 4. In expensive**

Pulse-Code Modulation

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- ❑ **ADC:** Convert analog signal to digital format

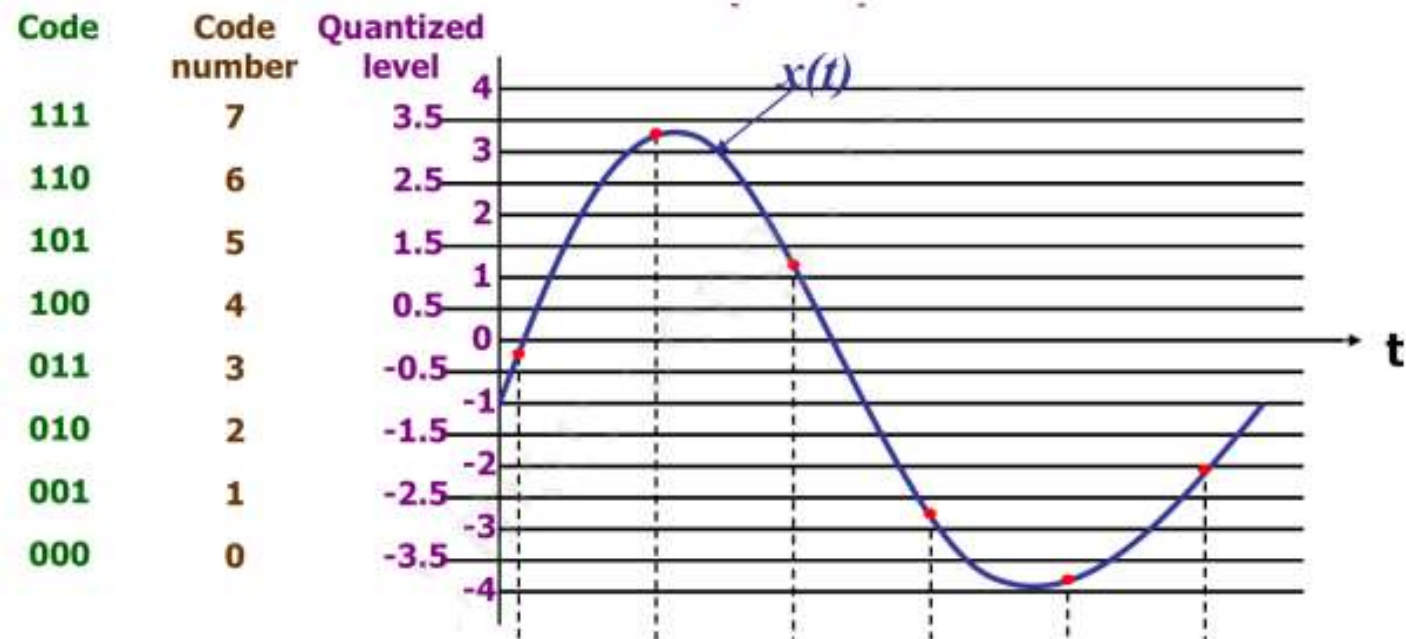


Sampling: a continuous-time signal is sampled by measuring its amplitude at discrete time instants.

Quantizing: represents the sampled values of the amplitude by a finite set of levels

Encoding: designates each quantized level by a digital code

Example:



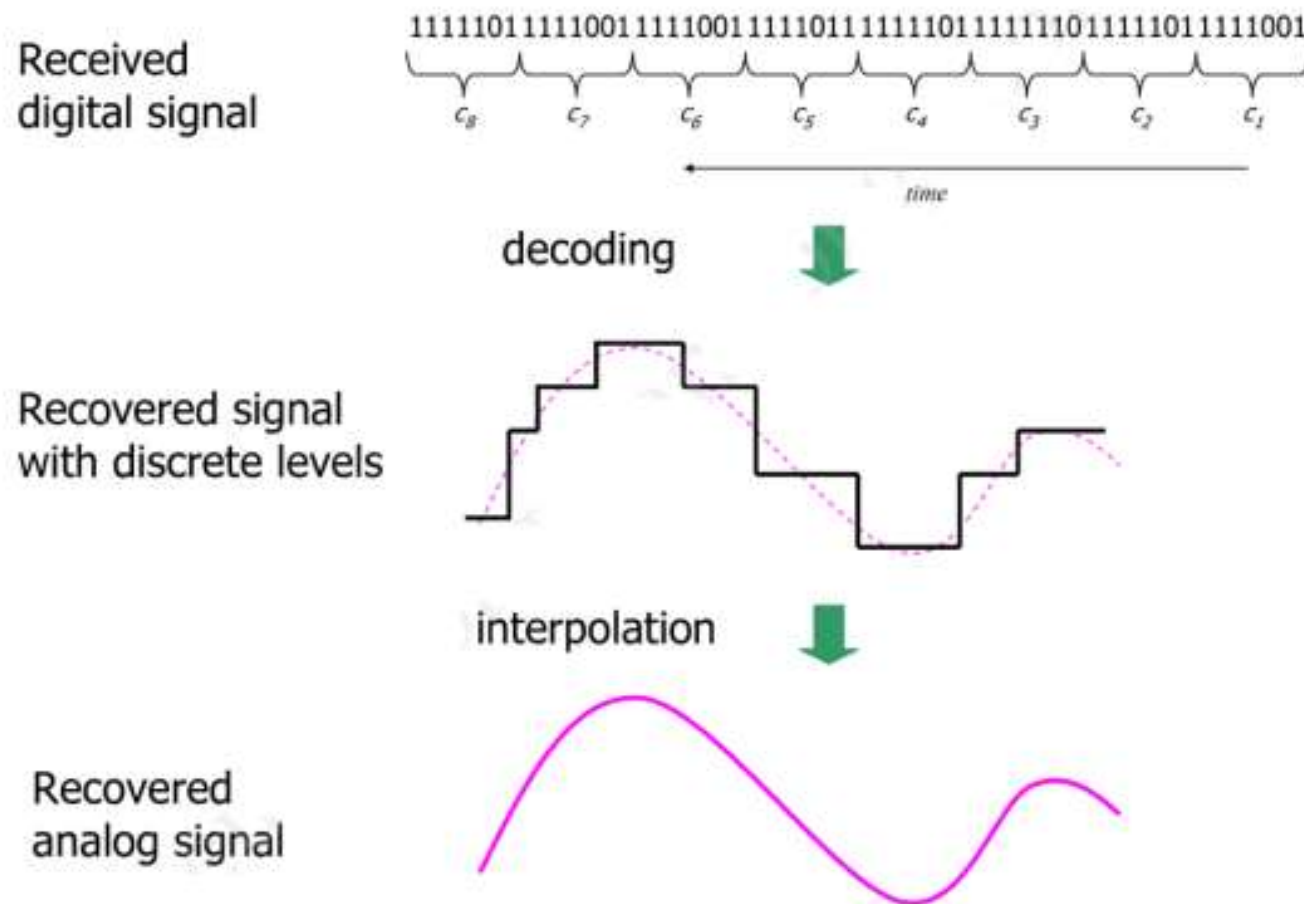
Sampler	Sampled value	-0.25	3.3	1.2	-2.8	-3.8	-2.1
Quantizer	Quantized value	-0.5	3.5	1.5	-2.5	-3.5	-2.5
Encoder	Code number	3	7	5	1	0	1
	Binary code	011	111	101	001	000	001
Parallel-to-serial converter							

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Pulse-Code Modulation

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- ❑ **DAC**: Reconstruct the digital signal to analog signal



Pulse-Code Modulation

The **encoder** works with M -ary digits and produces for each sample a **code word** of ν digits. If the quantizer has q **quantum levels**, then:

$$q = M^\nu \quad \nu = \log_M q$$

For binary PCM, $q = 2^\nu$.

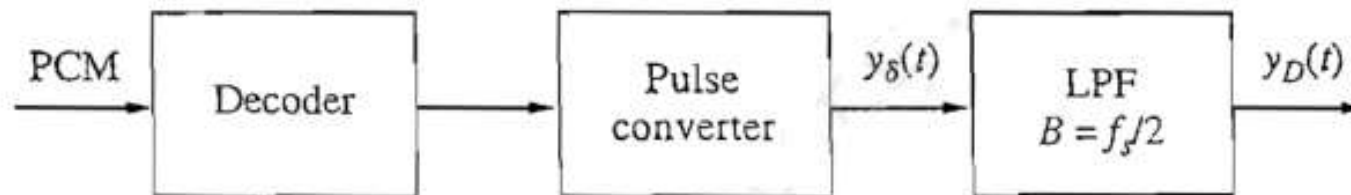
The signaling rate becomes: $r = \nu f_s$, where $f_s \geq 2W$.

Therefore, the bandwidth needed for PCM baseband transmission is:

$$B_T \geq \frac{1}{2}r = \frac{1}{2}\nu f_s \geq \nu W$$

□ Quantization Noise:

For convenient analysis of quantization noise, we use the impulse reconstruction model below:



The signal at output of the pulse converter is:

$$y_\delta(t) = \sum_k [x(kT_s) + \epsilon_k] \delta(t - kT_s)$$

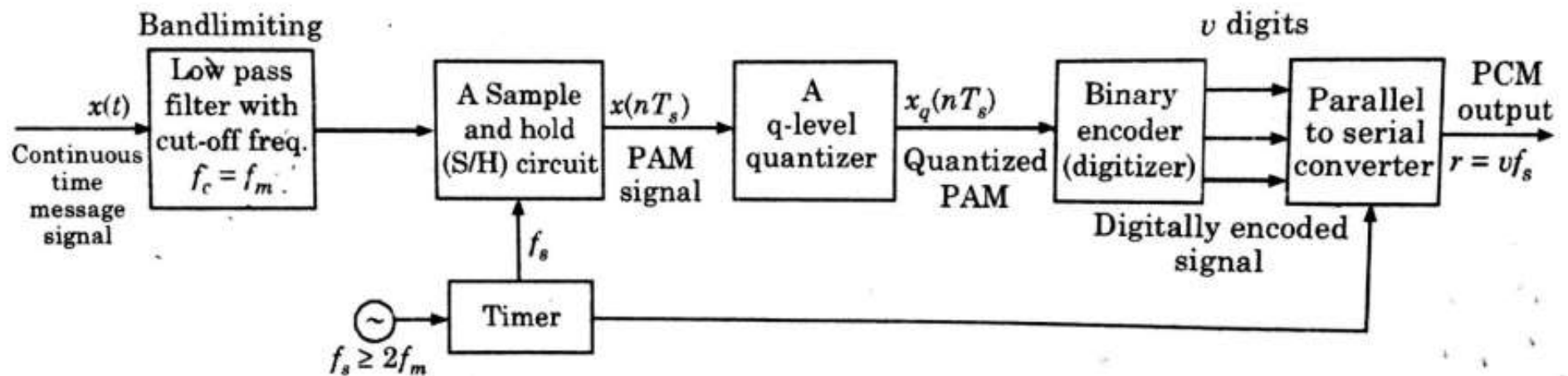
where ϵ_k represent the **quantization error** (or **quantization noise**):

$$\epsilon_k = x_q(kT_s) - x(kT_s)$$

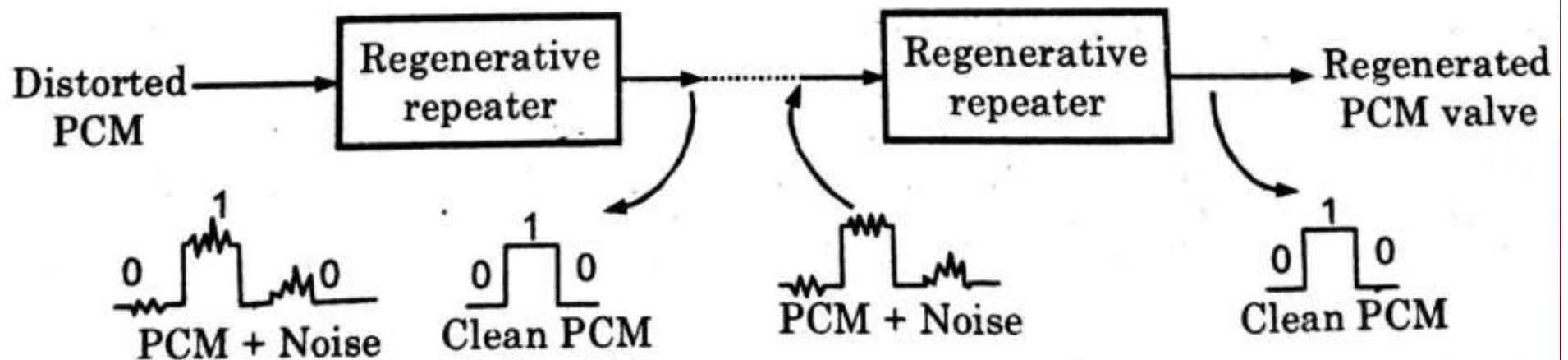
LPF with bandwidth $B = f_s/2$ yields:

$$y_D(t) = x(t) + \sum_k \epsilon_k \text{sinc}(f_s t - k)$$

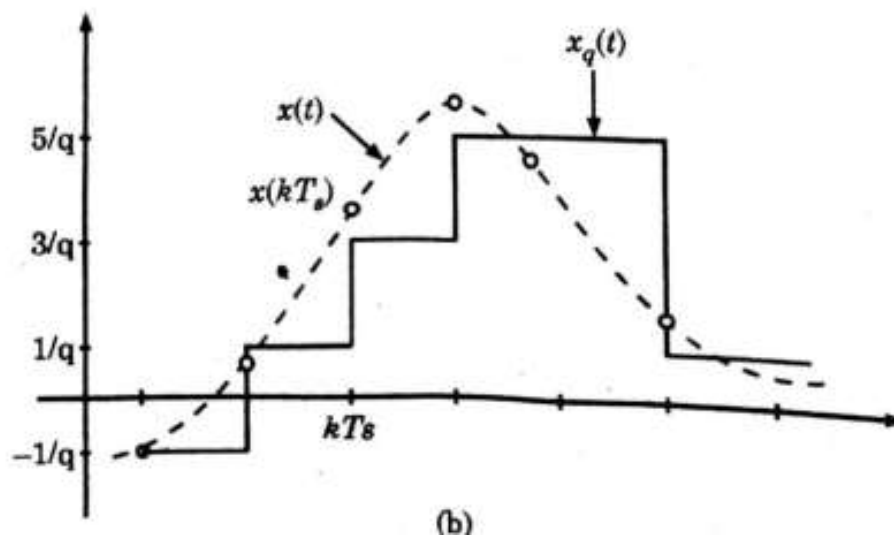
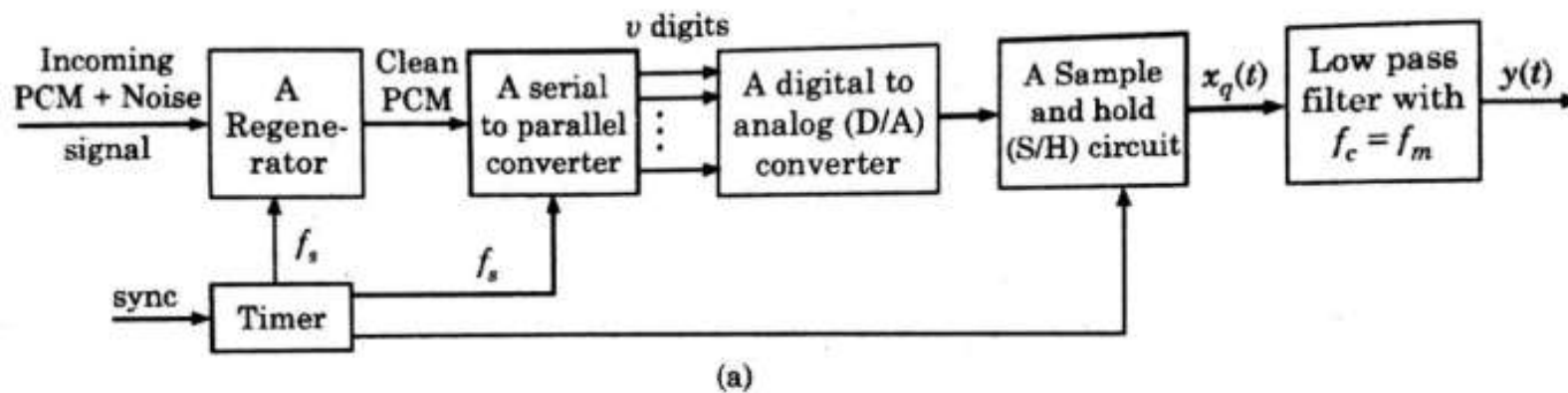
PCM Transmitter



PCM Transmission Path



PCM Receiver



Bandwidth of PCM

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Assume $w(t)$ is band limited to B hertz.

Minimum sampling rate = $2B$ samples / second

A/D output = n bits per sample (quantization level $M=2^n$)

Assume a simple PCM without redundancy.

Minimum channel bandwidth = bit rate / 2

➤ Bandwidth of PCM signals:

$B_{\text{PCM}} \geq nB$ (with sinc functions as orthogonal basis)

$B_{\text{PCM}} \geq 2nB$ (with rectangular pulses as orthogonal basis)

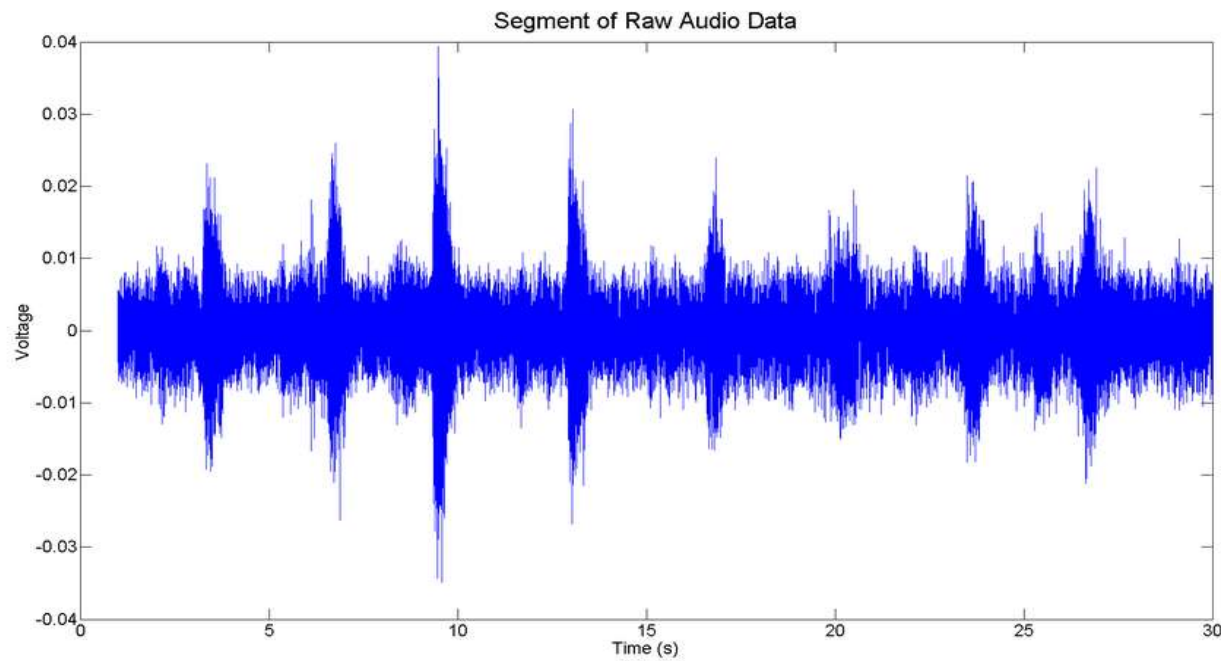
➤ For any reasonable quantization level M , PCM requires much higher bandwidth than the original $w(t)$.

II. PCM (cont.)

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- Linear Quantization:

$$q = \frac{S_{max} - S_{min}}{L} \quad \text{Với} \quad L = 2^n - 1$$



II. PCM (cont.)

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- Nonlinear Quantization:

North American standard:

$$y = \text{sgn}(x) \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)}$$

With $\mu = 255$ and

$$x = \frac{S}{V_{\max_input\ scale}}$$
$$\text{sgn}(x) = \begin{cases} -1 & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 & \text{if } x > 0 \end{cases}$$

European standard:

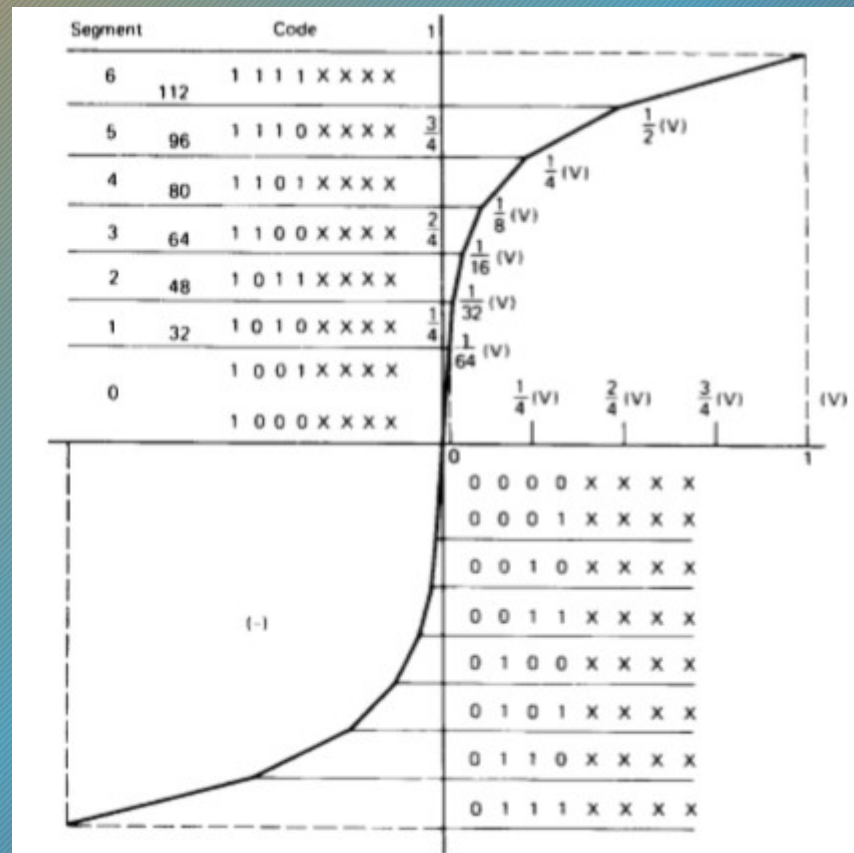
$$y = \text{sgn}(x) \frac{\ln(1 + A|x|)}{1 + \ln(A)} \quad \text{With } \frac{1}{A} \leq |x| \leq 1$$

$$y = \text{sgn}(x) \frac{A|x|}{1 + \ln(A)} \quad \text{With } 0 \leq |x| \leq \frac{1}{A}$$

II. PCM (cont.)

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- Nonlinear Quantization:



A-law quantization

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Advantages of PCM

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Relatively inexpensive.

Easily multiplexed.

Easily regenerated.

Better noise performance than analog system.

Signals may be stored and time-scaled efficiently.

Efficient codes are readily available.

Disadvantage

Requires wider bandwidth than analog signals

II. Delta Modulation

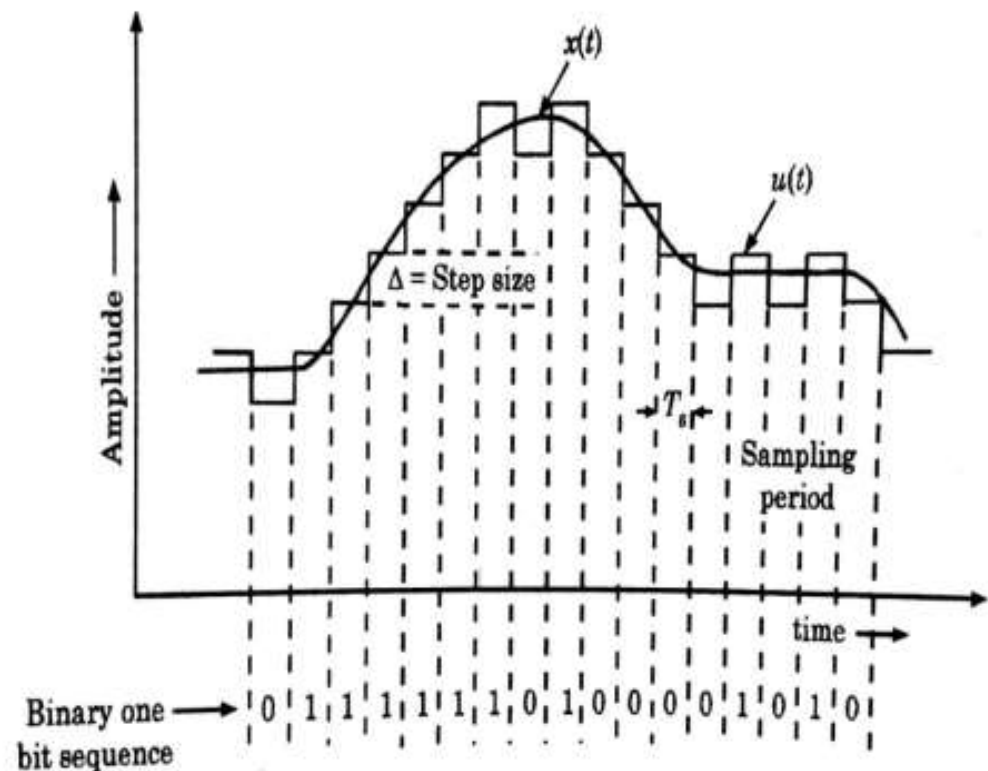
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- This scheme sends only the difference between pulses, if the pulse at time t_{n+1} is higher in amplitude value than the pulse at time t_n , then a single bit, say a “1”, is used to indicate the positive value.
- If the pulse is lower in value, resulting in a negative value, a “0” is used.
- This scheme works well for small changes in signal values between samples.
- If changes in amplitude are large, this will result in large errors.

Delta Modulation

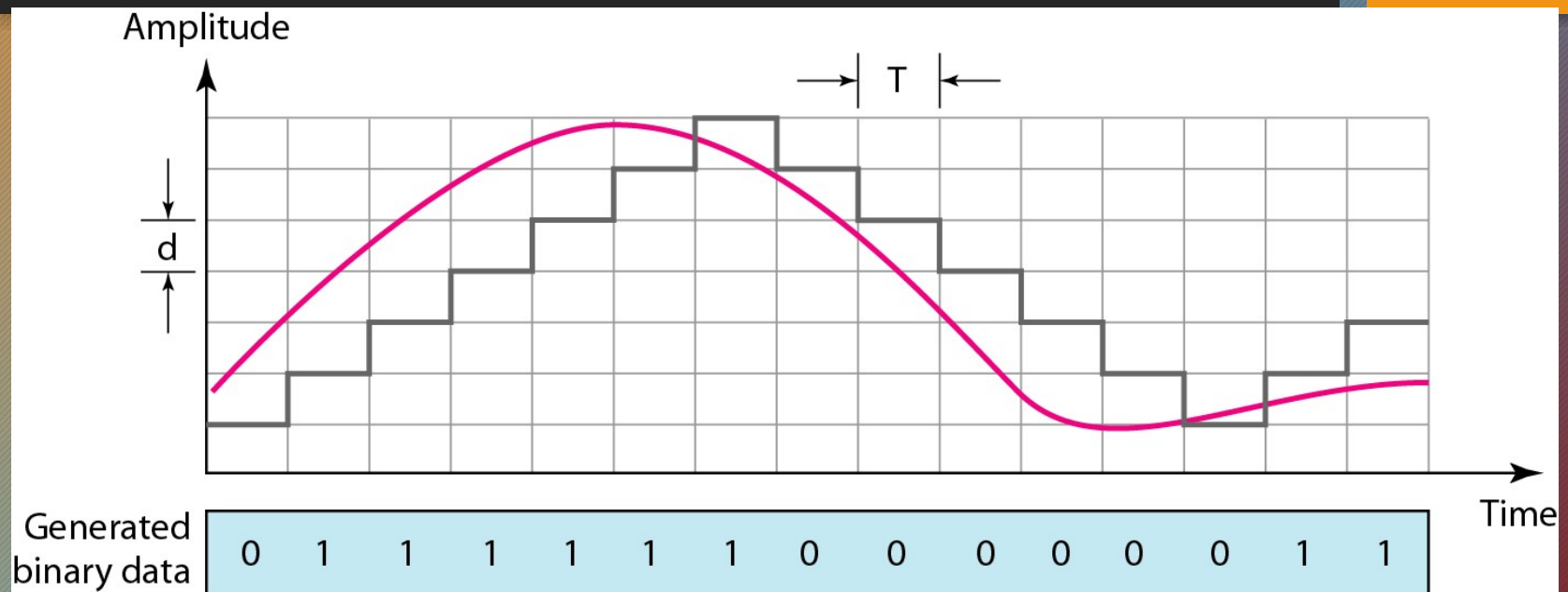
Types of noise

- Quantization noise: step size δ takes place of smallest quantization level.
- δ too small: slope overload noise
- too large: quantization noise and granular noise



II. Delta Modulation

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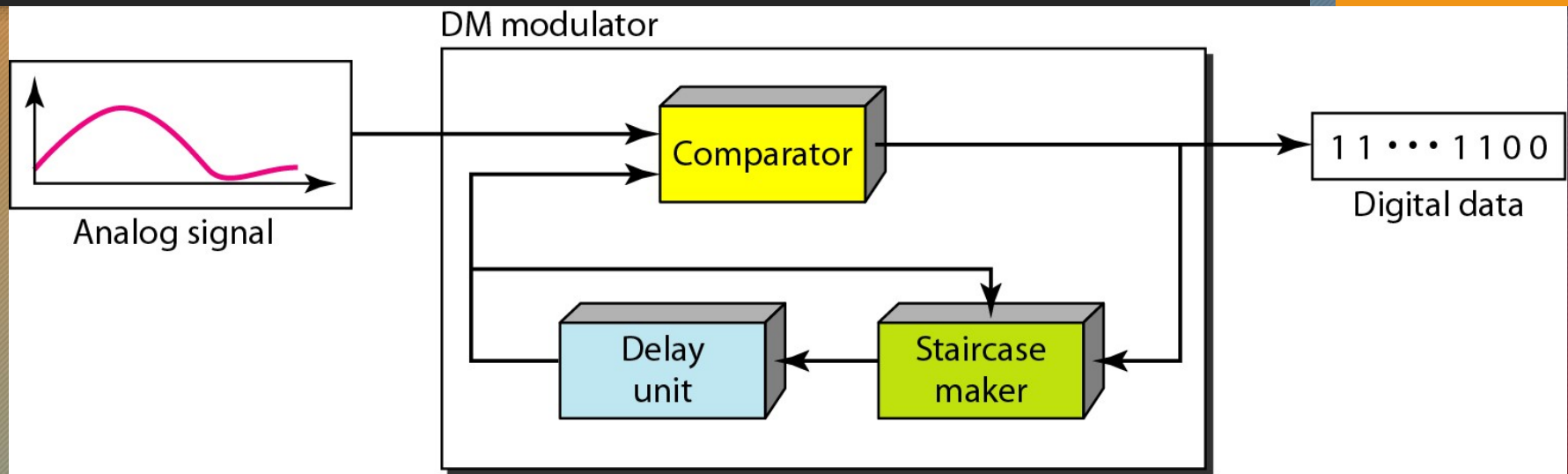


The process of delta modulation

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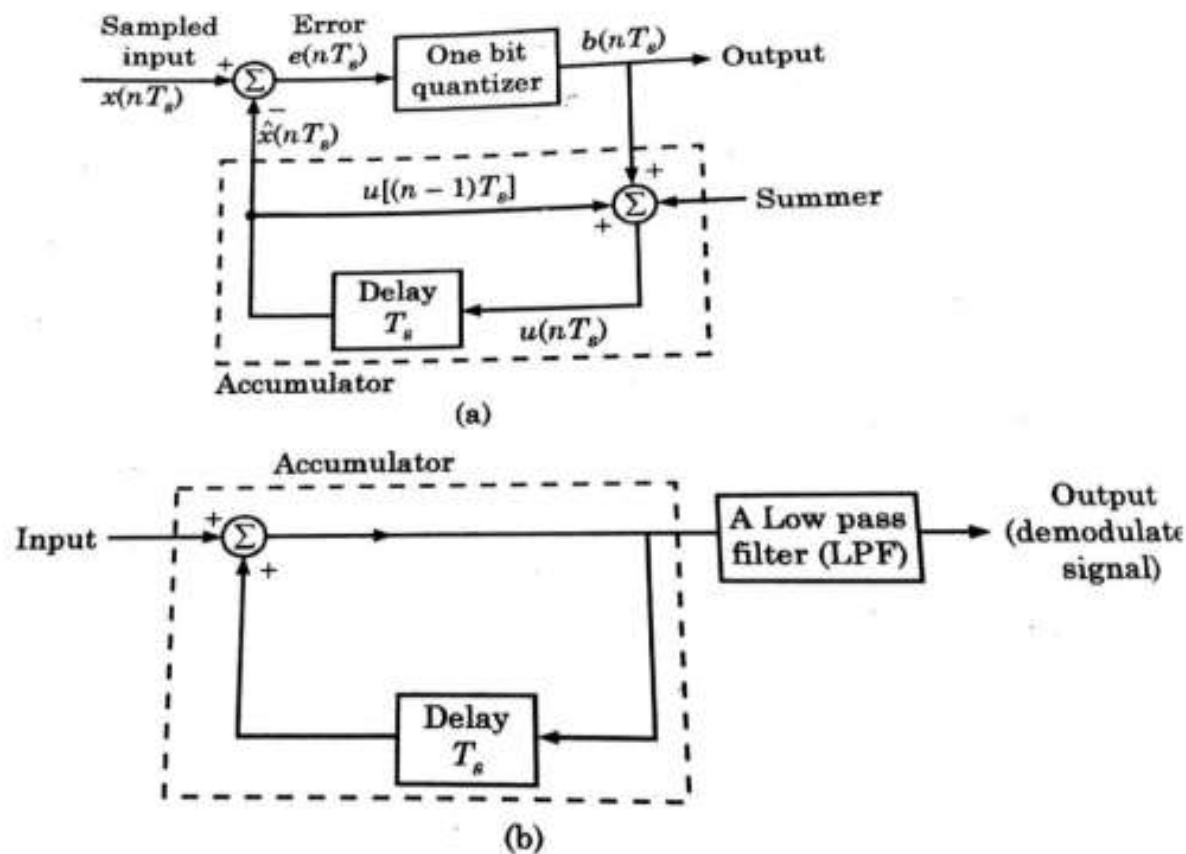
II. Delta Modulation (cont.)

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Delta modulation components

Delta Modulator Transmitter & Receiver



II. Delta Modulation

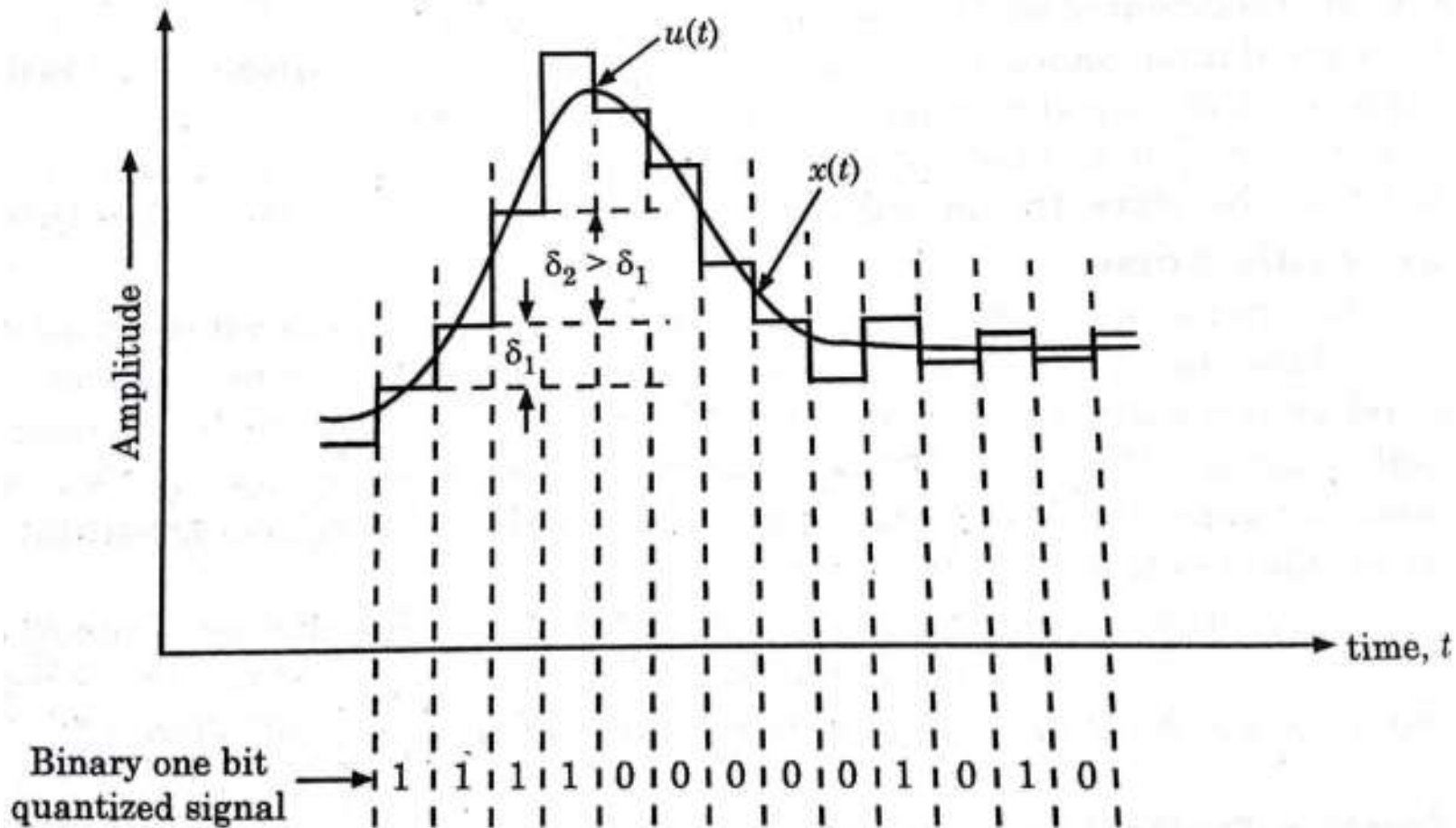
4.64

- Instead of using one bit to indicate positive and negative differences, we can use more bits → quantization of the difference.
- Each bit code is used to represent the value of the difference.
- The more bits the more levels -> the higher the accuracy.

ADM

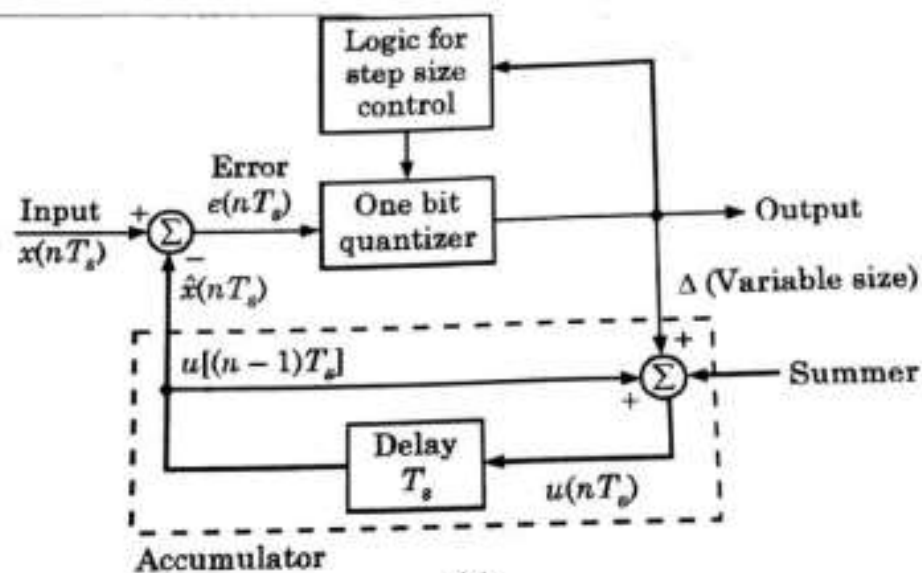
Adaptive Delta Modulation

4.65

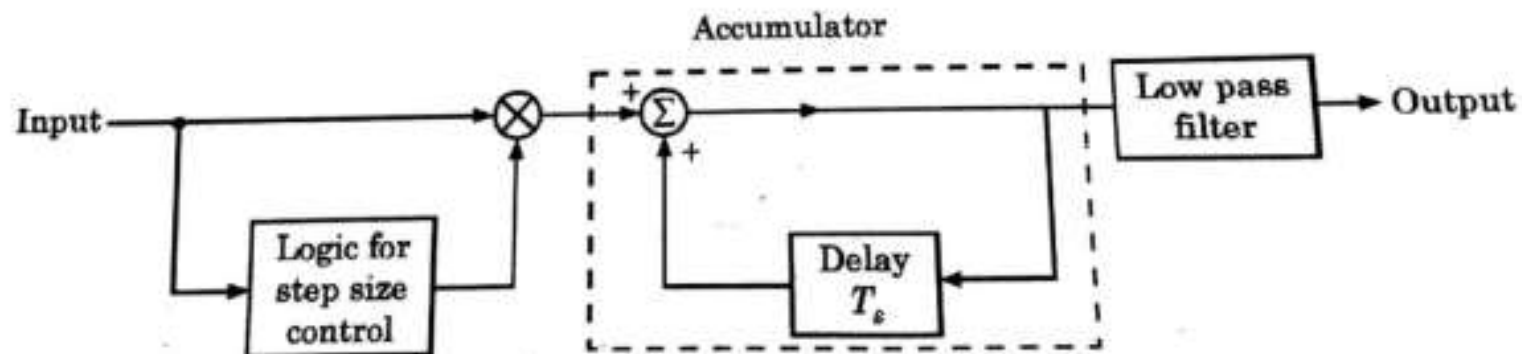


ADM Transmitter & Receiver

4.66



(a)



(b)

DPCM

Differential Pulse Code Modulation

4.67

Often voice and video signals do not change much from one sample to next.

- Such signals has energy concentrated in lower frequency.

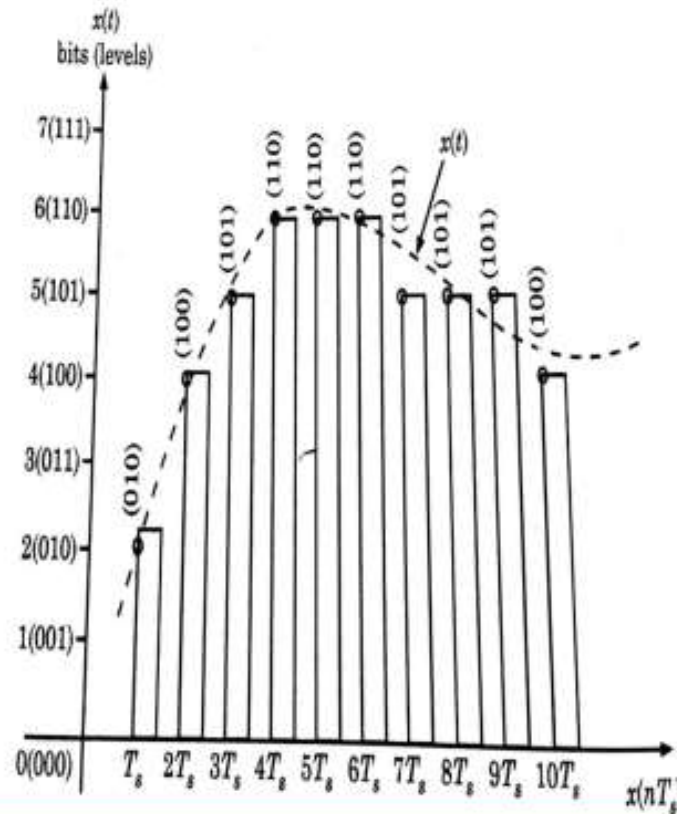
- Sampling faster than necessary generates redundant information.

Can save bandwidth by not sending all samples.

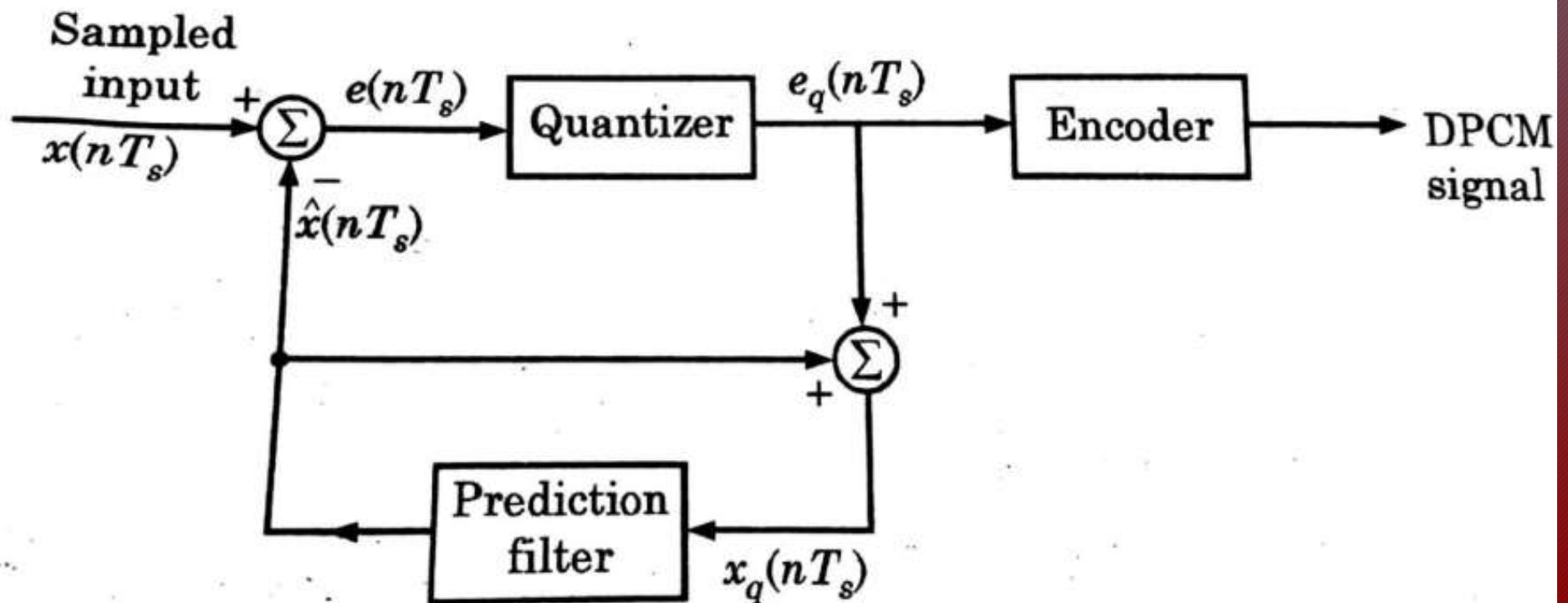
- * Send true samples occasionally.

- * In between, send only change from previous value.

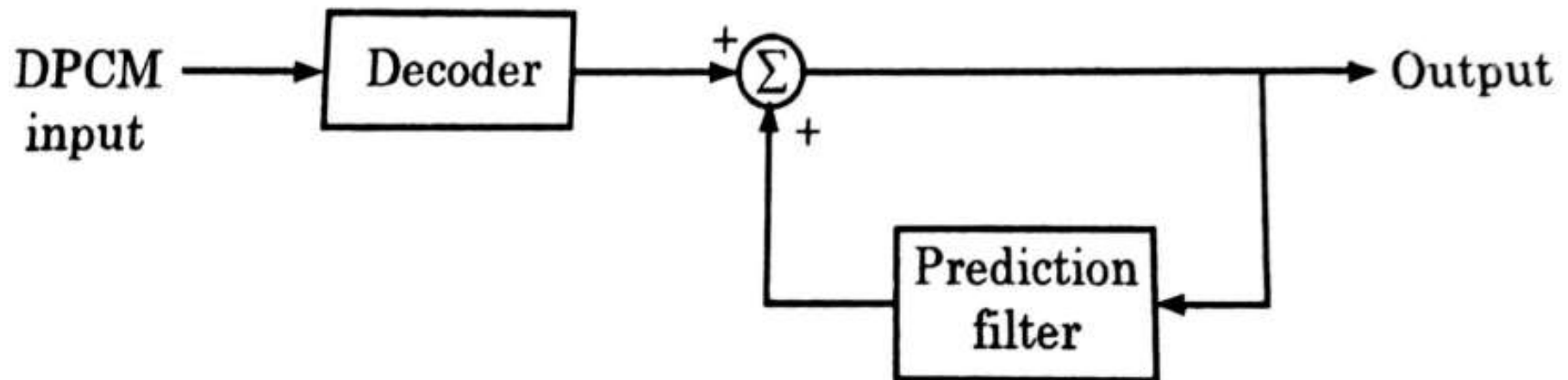
- * Change values can be sent using a fewer number of bits than true samples.



DPCM Transmitter



DPCM Receiver



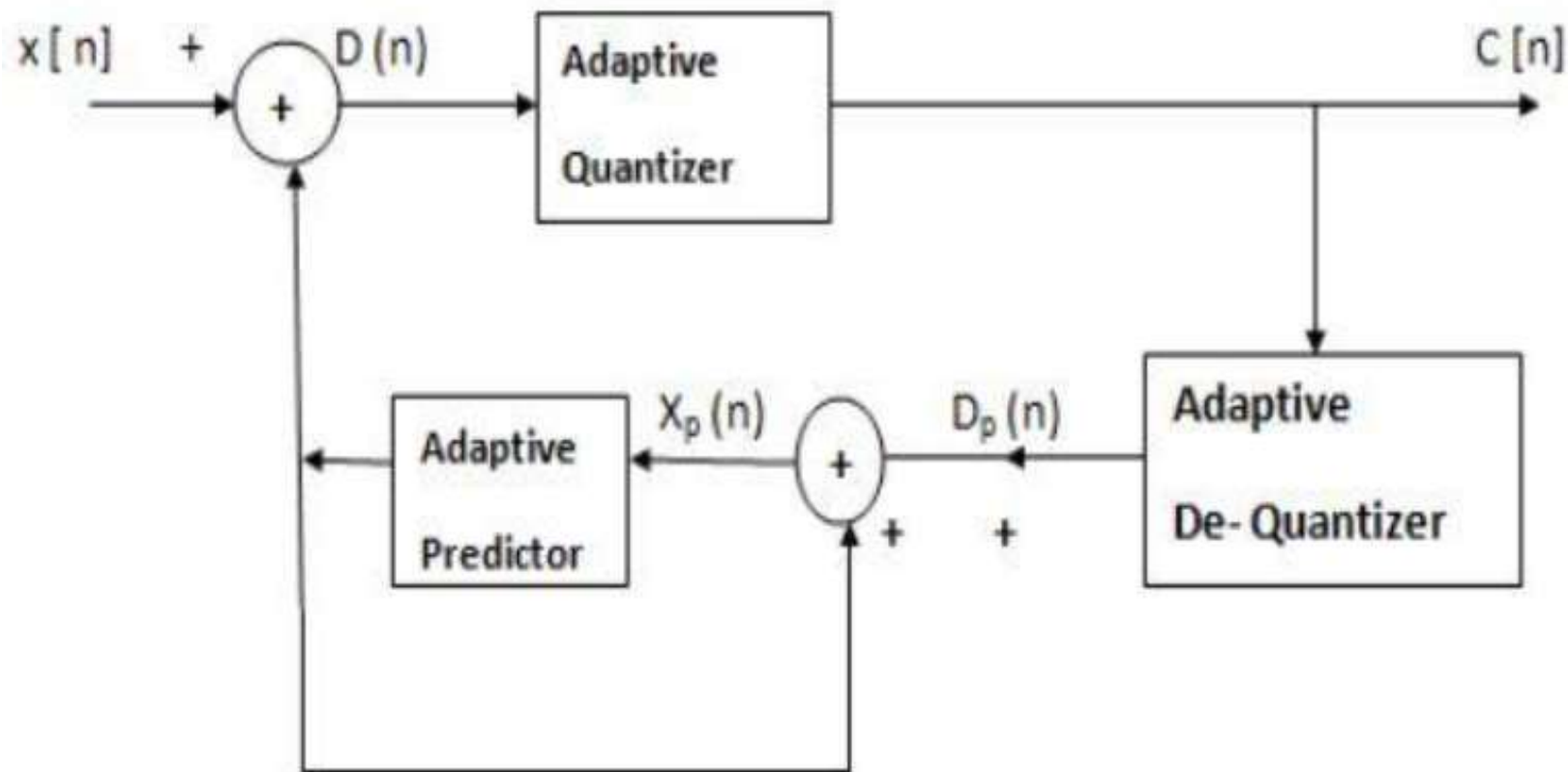
Processing Gain

In a spread-spectrum system, the process gain (or "processing gain") is the ratio of the spread (or RF) bandwidth to the unspread (or baseband) bandwidth. It is usually expressed in decibels (dB).

For example, if a 1 kHz signal is spread to 100 kHz, the process gain expressed as a numerical ratio would be $100000/1000 = 100$. Or in decibels, $10 \log_{10}(100) = 20$ dB.

4.71

Adaptive DPCM

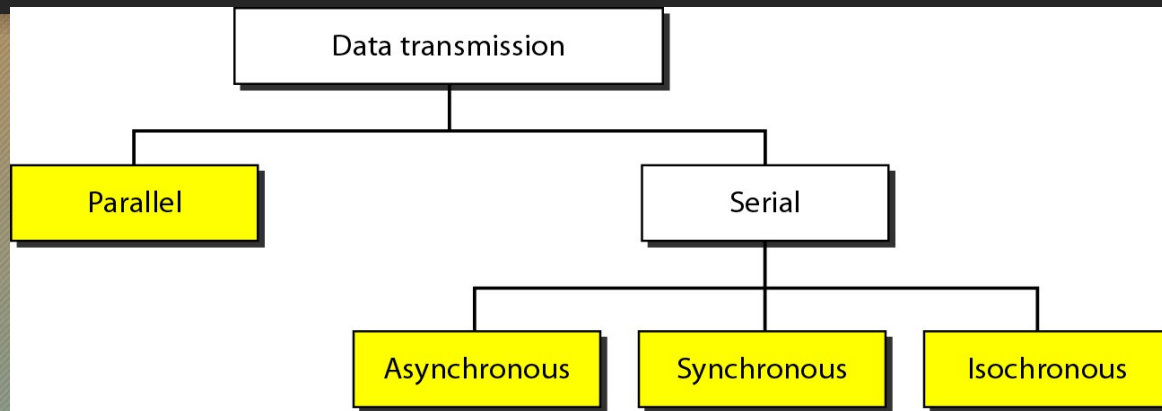


Comparisons

<i>S. No.</i>	<i>Parameter of comparison</i>	<i>Pulse Code Modulation (PCM)</i>	<i>Delta modulation (DM)</i>	<i>Adaptive Delta Modulation (ADM)</i>	<i>Differential Pulse Code Modulation (DPCM)</i>
1.	Number of bits.	It can use 4, 8 or 16 bits per sample.	It uses only one bit for one sample.	Only one bit is used to encode one sample.	Bits can be more than one but are less than PCM.
2.	Levels and step size	The number of levels depend on number of bits. Level size is kept fixed.	Step size is kept fixed and cannot be varied.	According to the signal variation, step size varies (i.e. Adapted).	Here, Fixed number of levels are used.
3.	Quantization error and distortion	Quantization error depends on number of levels used.	Slope overload distortion and granular noise are present.	Quantization noise is present but other errors are absent.	Slope overload distortion and quantization noise is present.
4.	Transmission bandwidth	Highest bandwidth is required since number of bits are high	Lowest bandwidth is required.	Lowest bandwidth is required.	Bandwidth required is lower than PCM.
5.	Feedback	There is no feedback in transmitter or receiver.	Feedback exists in transmitter.	Feedback exists.	Here, Feedback exists.
6.	Complexity of implementation	System complex.	Simple.	Simple.	Simple

III. Digital transmission modes

4.73

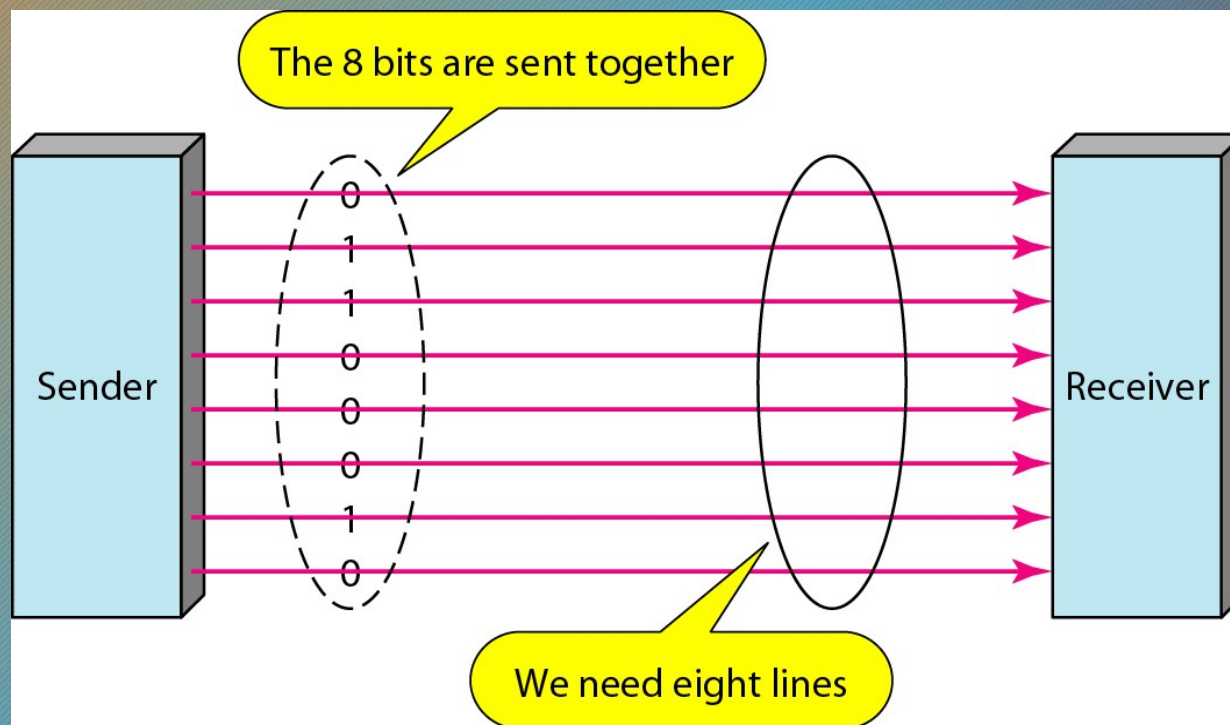


- **Parallel** transmission is sending multiple bits at the same time over separate media
- **Serial** transmission is sending one bit at a time
 - **Asynchronous:** This is transmission at any time, with arbitrary delay between transmission of any two successive data items.
 - **Synchronous:** This is continuous transmission with no gaps between transmission of successive data items.
 - **Isochronous:** This is transmission at regular intervals with a fixed gap between the transmission of successive data items.

III. Digital transmission modes (cont.)

4.74

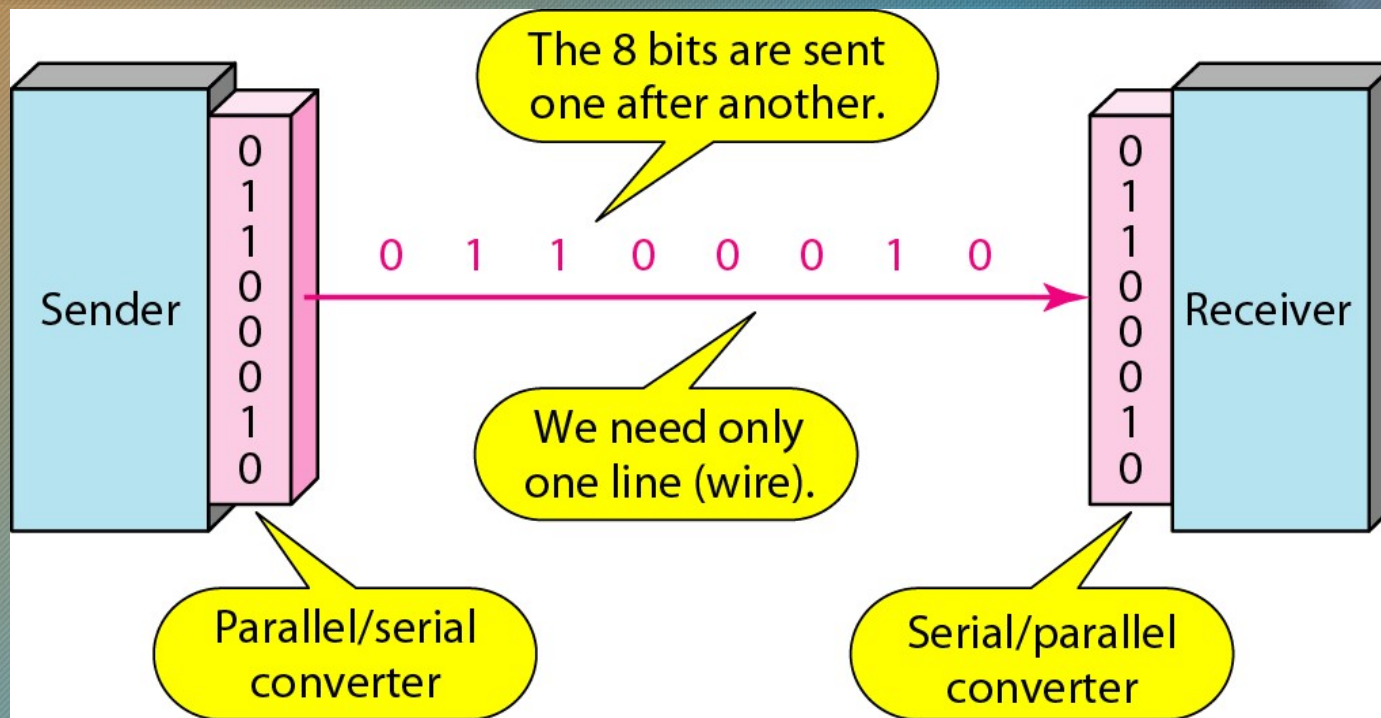
Parallel transmission



III. Digital transmission modes (cont.)

4.75

Serial transmission



Serial transmission

III. Digital transmission modes (cont.)

4.76

Note

file: 200kb $\Rightarrow$

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte. There may be a gap between each byte.

Why we has to do this?

8 bit data $\rightarrow 10$
200k $\rightarrow 200$
 $R = 9600$
 $\Rightarrow t = \frac{S}{R}$

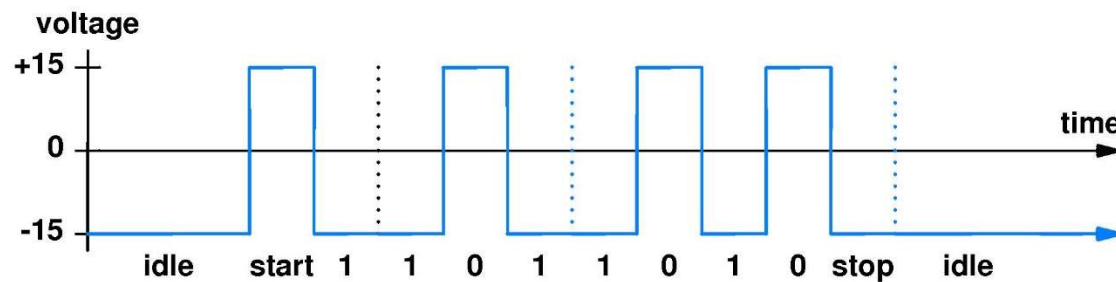
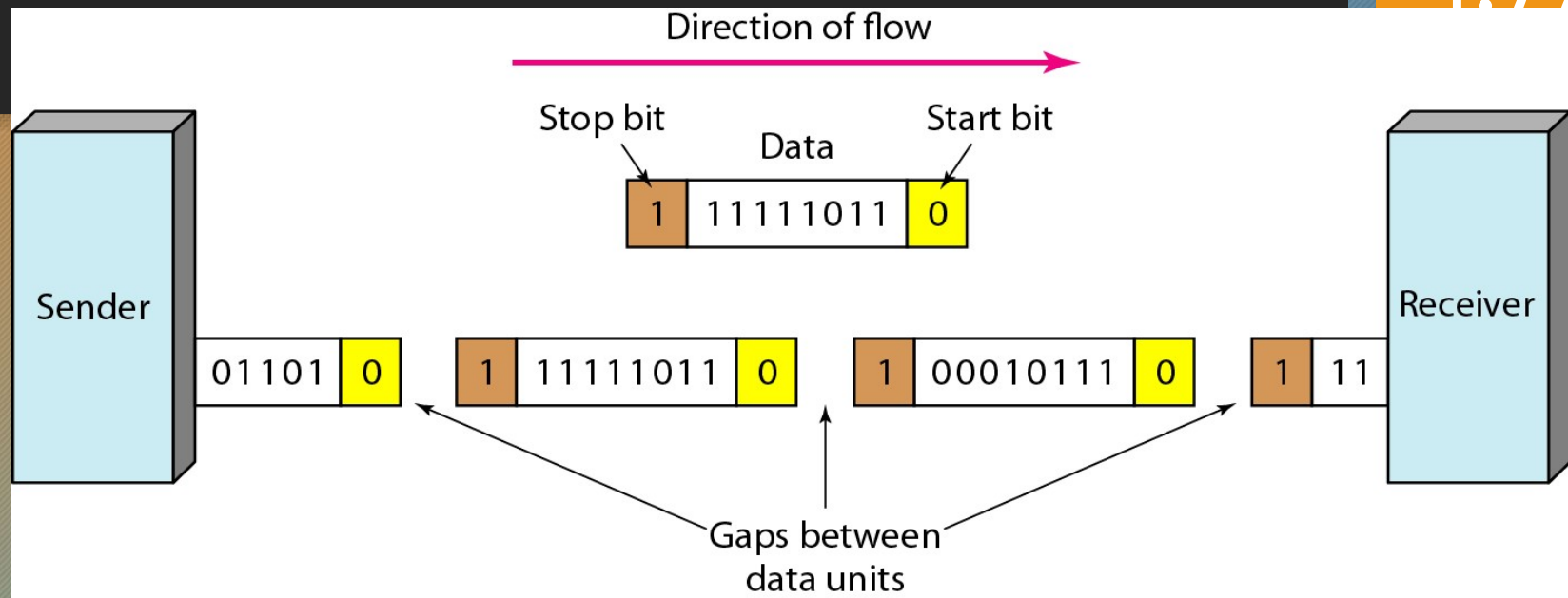


Figure 9.5 Illustration of voltage during transmission of an 8-bit character when using RS-232.

III. Digital Transmission modes (cont.)

4.77



Asynchronous transmission

A PCM transmission system has a low-pass analog signal input of 4 kHz, sampling 8 bits, asynchronous serial transmission with the character format: 1 start bit/2 stop bits.

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What is the bit-rate of the transmission system?

III. Digital Transmission modes (cont.)

Note

In synchronous transmission, we send bits one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits. The bits are usually sent as bytes and many bytes are grouped in a frame. A frame is identified with **a start and an end byte**.

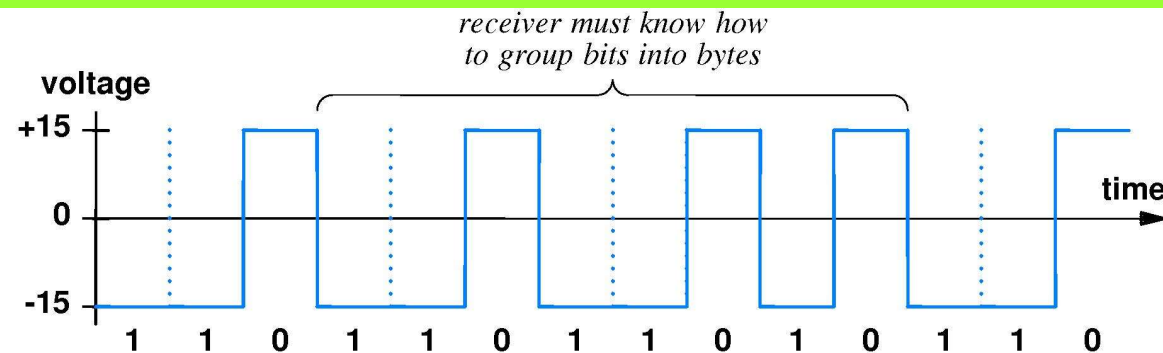
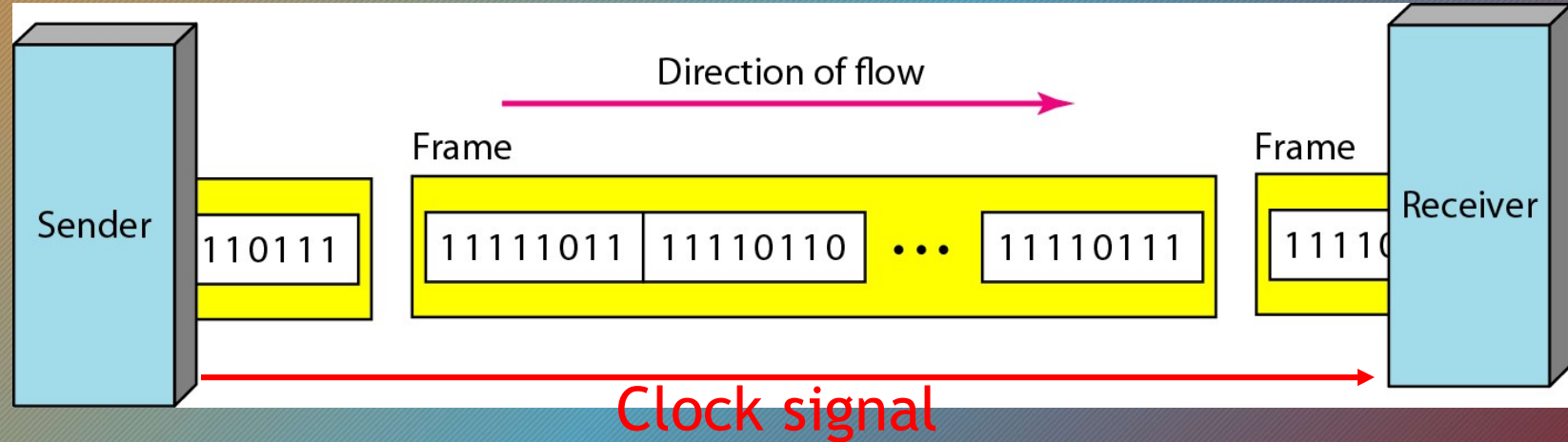


Figure 9.6 Illustration of synchronous transmission where the first bit of a byte immediately follows the last bit of the previous byte.

III. Digital Transmission modes (cont.)

4.79



Synchronous transmission

III. Digital Transmission modes (cont.)

4.80

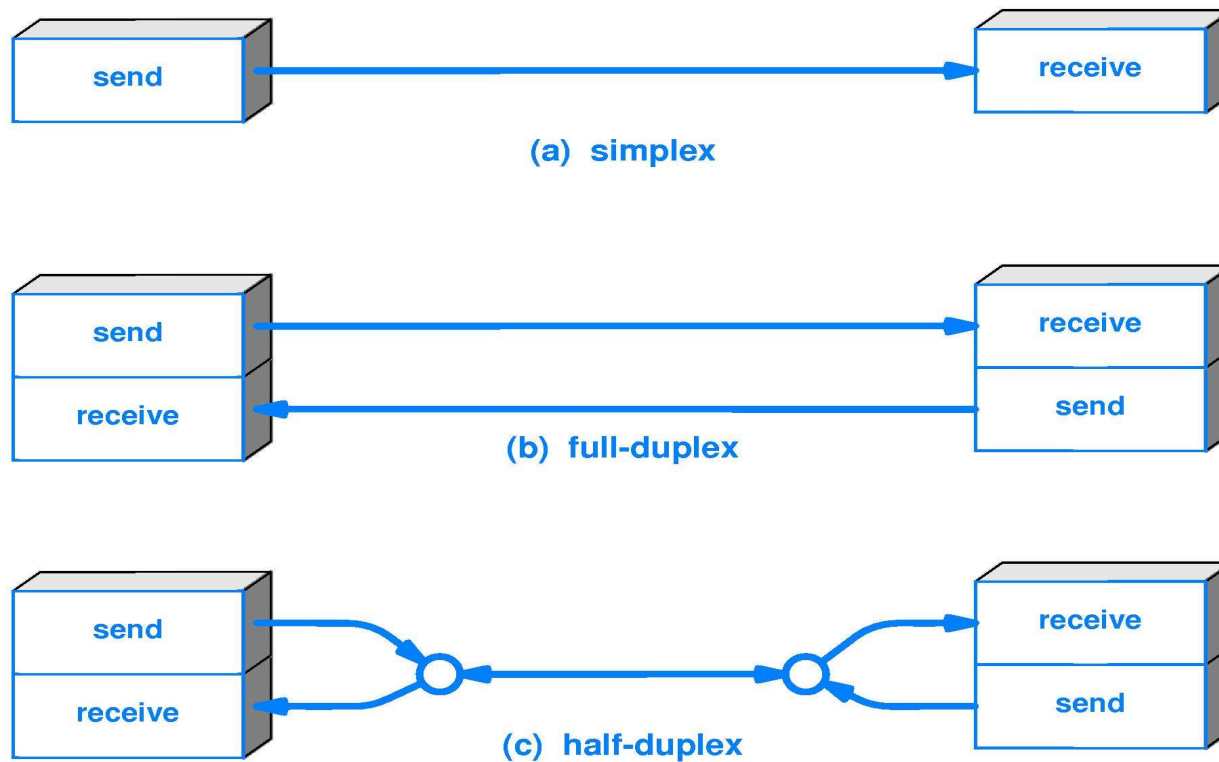
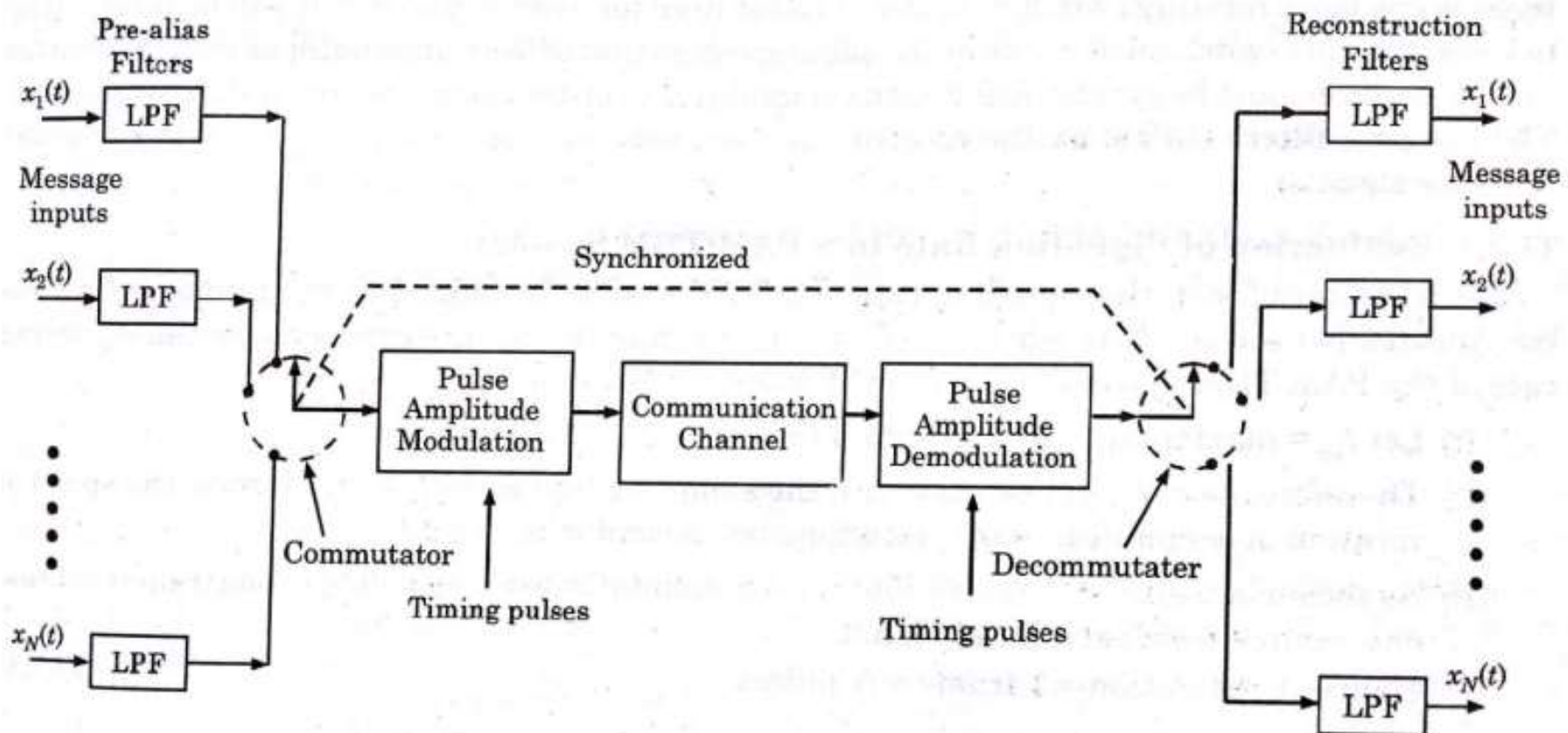
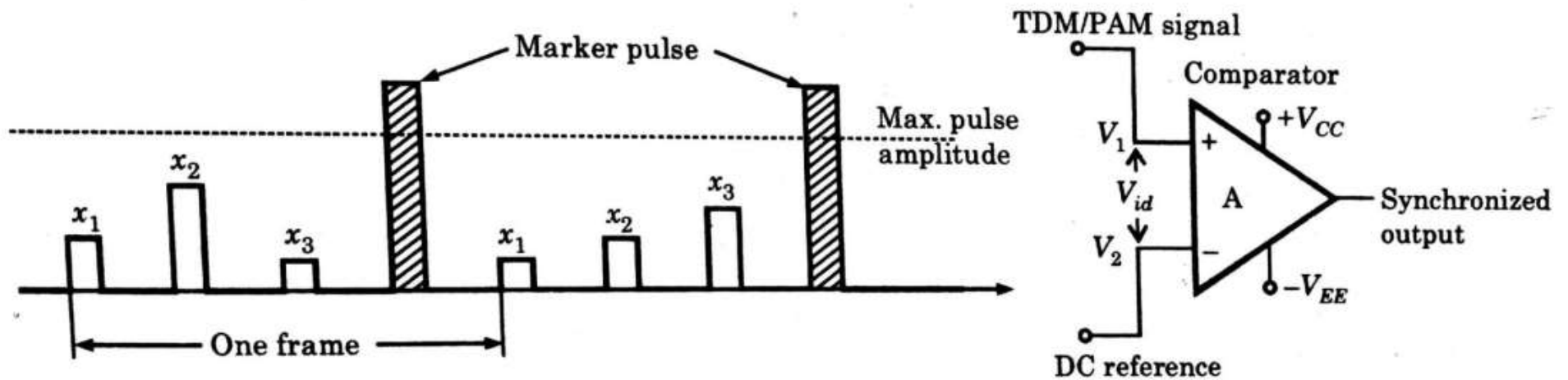


Figure 9.8 Illustration of the three modes of operation.

Time Division Multiplexing(TDM)

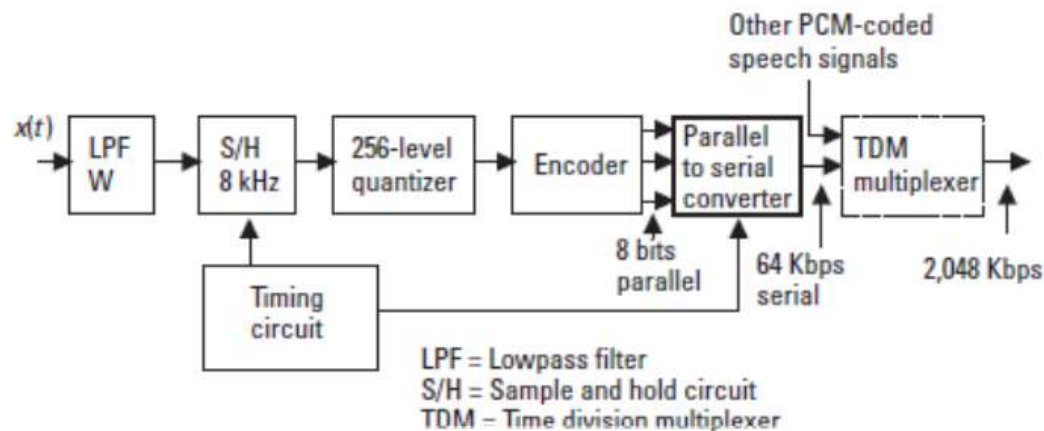


Synchronization



IV. TDM for E1/T1 system

4.83

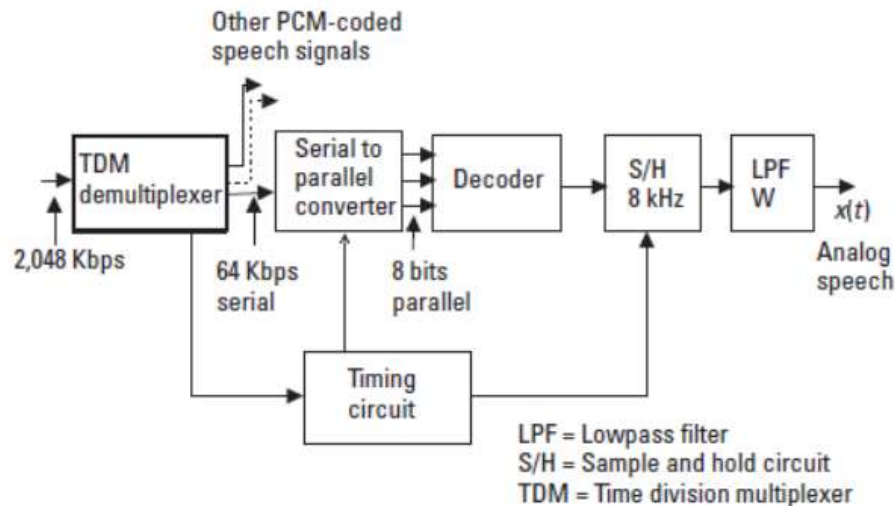


PCM encoder

Data output

European standard,
E1 = $32 \times 64 \text{ kbps} = 2,048 \text{ kbps}$

North American standard,
T1: $24 \times 64 \text{ kbps} = 1,536 \text{ kbps}$

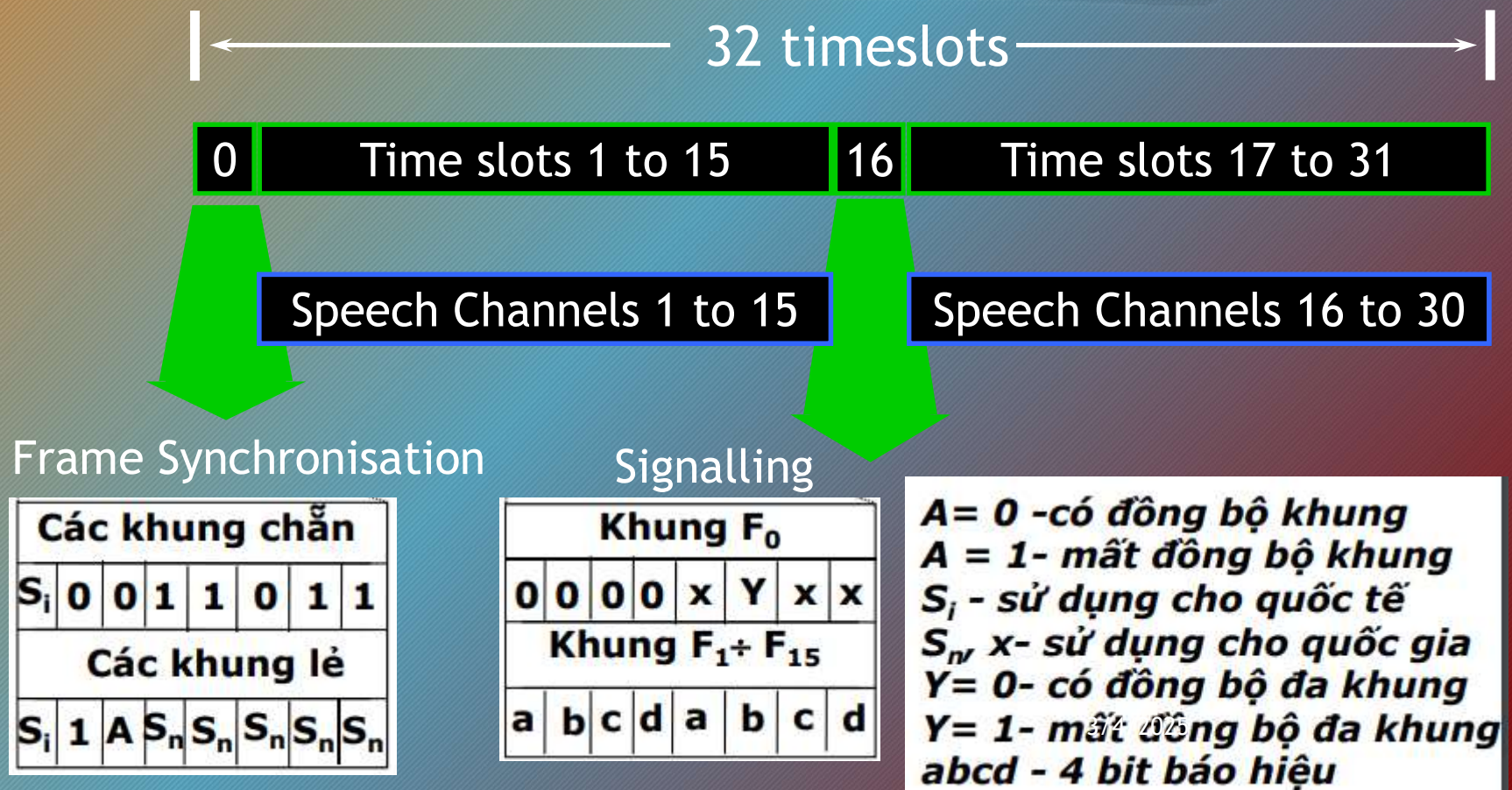


PCM decoder

4/2025

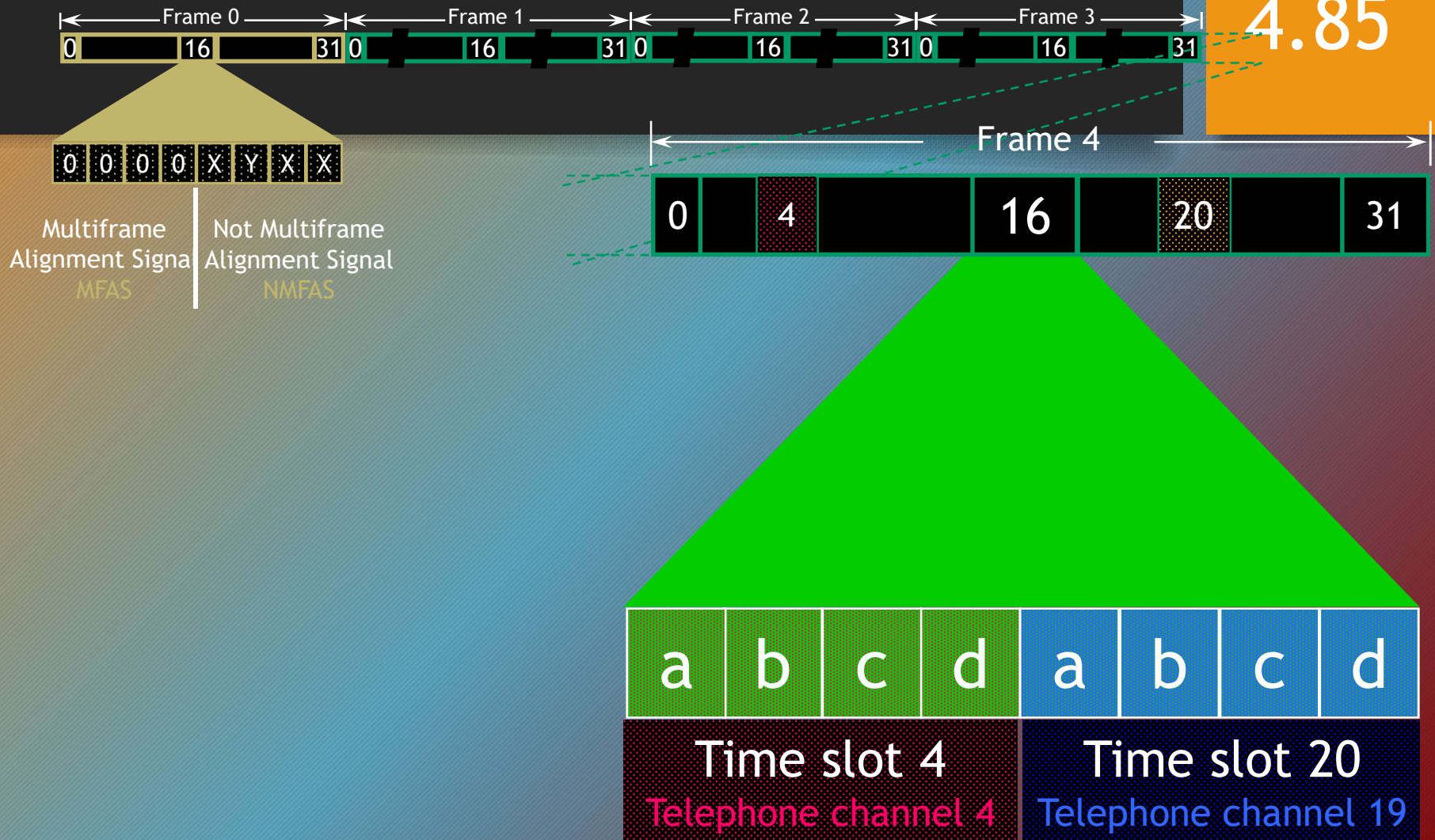
IV. TDM for E1/T1 system (cont.) 4.84

Data bit stream in a frame of E1



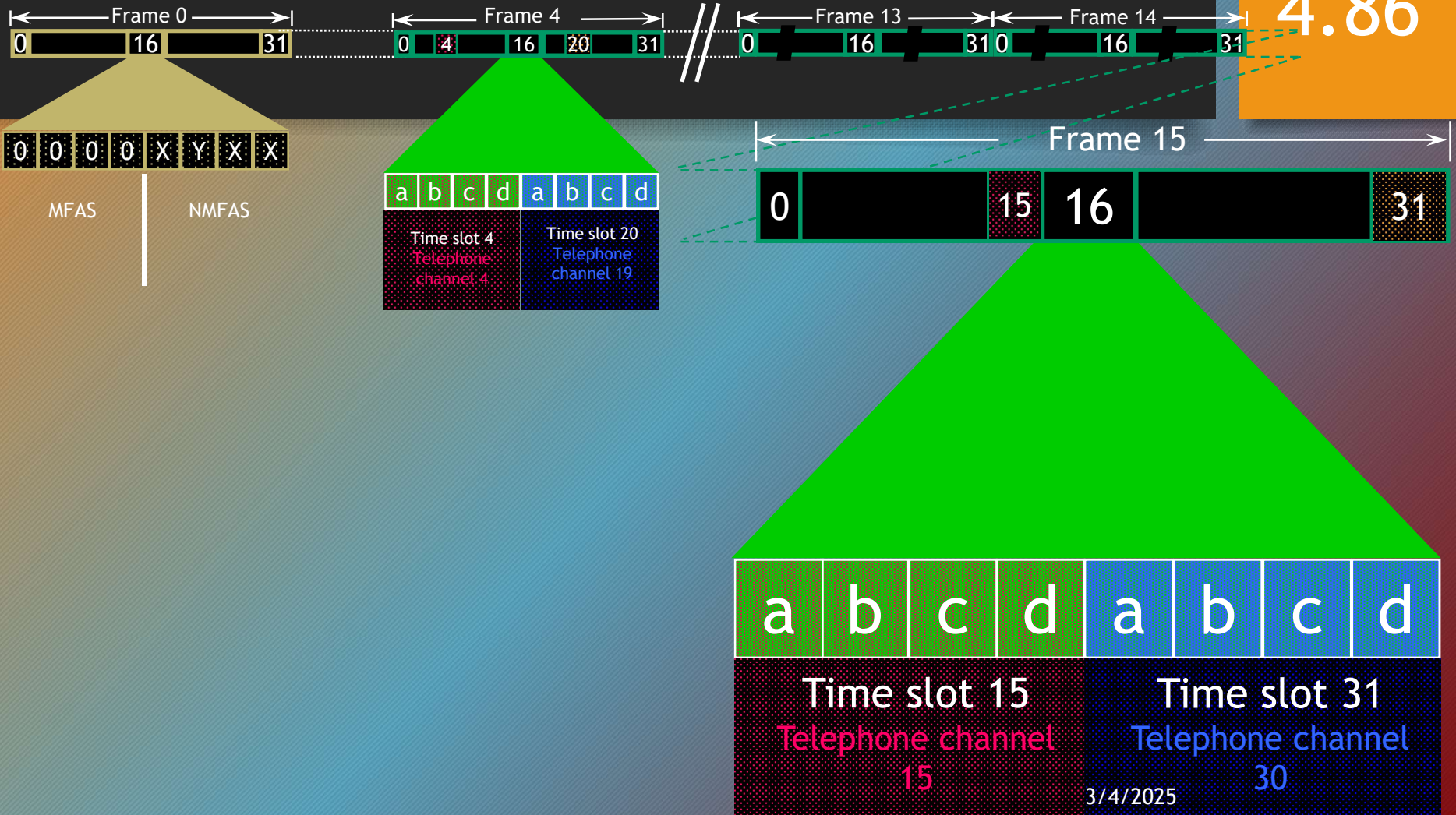
Signalling Multiframe

4.85



Signalling Multiframe

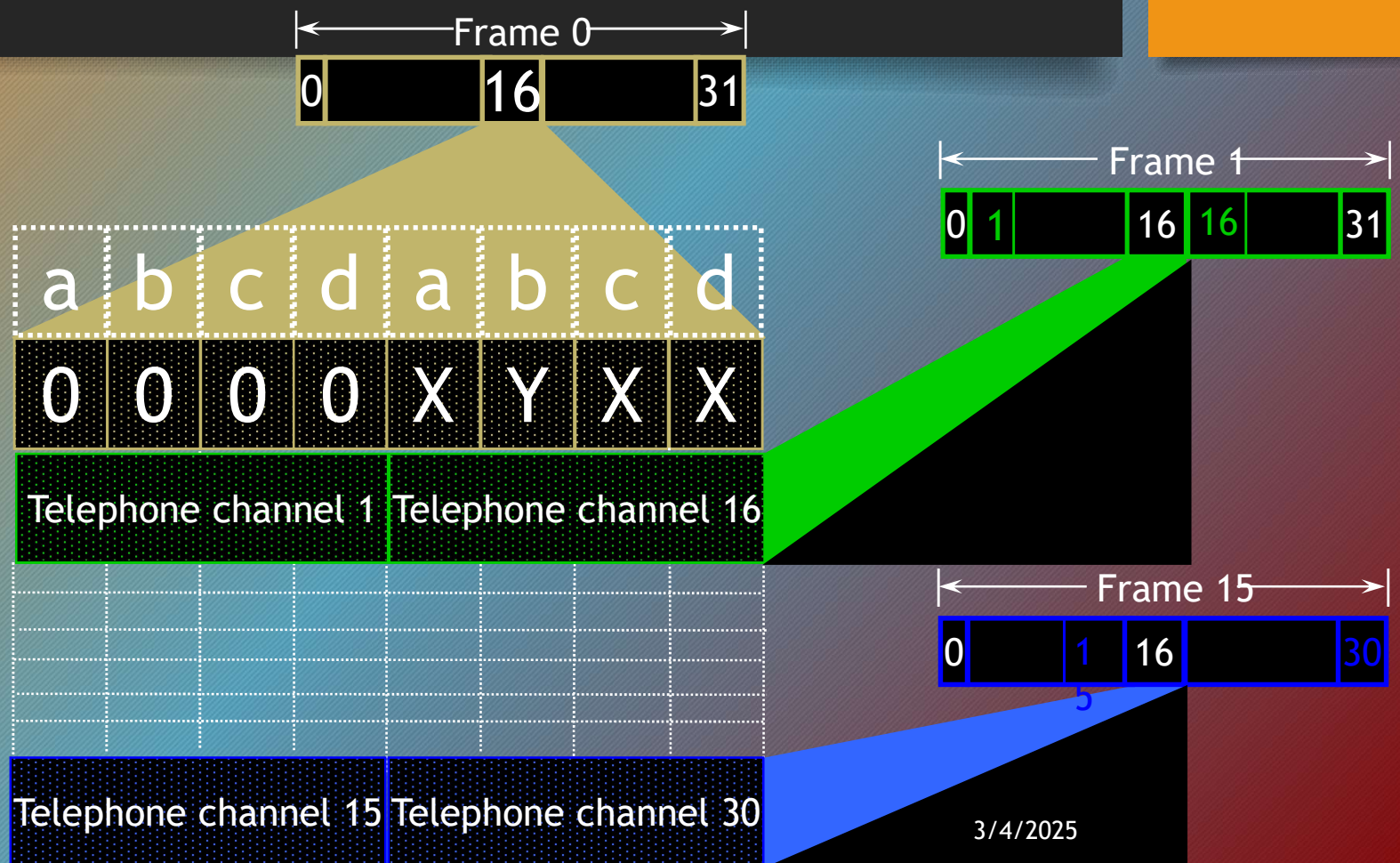
4.86



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Signalling Multiframe in E1 system

4.87



CAS Signalling in E1 system

4.88

PCM frame number	Bits in time slot 16							
	a	b	c	d	a	b	c	d
0	0	0	0	0	X	Y	X	X
1	Telephone channel 1				Telephone channel 16			
2	Telephone channel 2				Telephone channel 17			
3	Telephone channel 3				Telephone channel 18			
4	Telephone channel 4				Telephone channel 19			
5	Telephone channel 5				Telephone channel 20			
6	Telephone channel 6				Telephone channel 21			
7	Telephone channel 7				Telephone channel 22			
8	Telephone channel 8				Telephone channel 23			
9	Telephone channel 9				Telephone channel 24			
10	Telephone channel 10				Telephone channel 25			
11	Telephone channel 11				Telephone channel 26			
12	Telephone channel 12				Telephone channel 27			
13	Telephone channel 13				Telephone channel 28			
14	Telephone channel 14				Telephone channel 29			
15	Telephone channel 15				Telephone channel 30			

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Cyclic Redundancy Check

CRC - 4 example

4.89

Data block $\times 2^4 \div$ Generator polynomial $x^4 + x + 1$

1 0 1 0 0 0 0 0

1 0 0 1 1

1 0 0 1 1

1 1 1 0 0

1 0 0 1 1

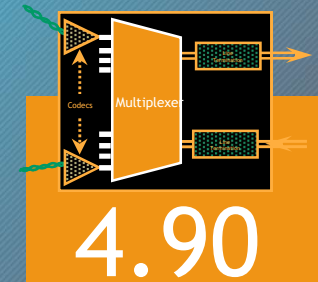
1 1 1 1 0

1 0 0 1 1

1 1 0 1

Remainder or *Signature*

CRC-4 bits



0 Time slots 1 to 15 16 Time slots 17 to 31

CRC check bits (remainder)

Frame Alignment
Signal (FAS)

8 bits

C _x	0	0	1	1	0	1	1
----------------	---	---	---	---	---	---	---

CRC Multiframe
Alignment Signal
001011

Not Frame
Alignment Signal
(NFAS)

X	1	A	S _{a4}	S _{a5}	S _{a6}	S _{a7}	S _{a8}
---	---	---	-----------------	-----------------	-----------------	-----------------	-----------------

CRC Multiframe

4.91

Sub-multiframe (SMF)	Frame number	Bits 1 to 8 of the frame							
		1	2	3	4	5	6	7	8
I	0	C_1	FAS						
	1	0	1	A	S_n	S_n	S_n	S_n	S_n
	2	C_2	FAS						
	3	0	1	A	S_n	S_n	S_n	S_n	S_n
	4	C_3	FAS						
	5	1	1	A	S_n	S_n	S_n	S_n	S_n
	6	C_4	FAS						
	7	0	1	A	S_n	S_n	S_n	S_n	S_n
II	8	C_1	FAS						
	9	1	1	A	S_n	S_n	S_n	S_n	S_n
	10	C_2	FAS						
	11	1	1	A	S_n	S_n	S_n	S_n	S_n
	12	C_3	FAS						
	13	E	1	A	S_n	S_n	S_n	S_n	S_n
	14	C_4	FAS						
	15	E	1	A	S_n	S_n	S_n	S_n	S_n

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T1 systems

4.92

- As mentioned previously, sampling rate used for voice = 8000 samples/sec
- Every sample is represented by 8 bits
- Therefore, data rate of 1 voice channel = $8 \times 8000 = 64\text{kbps}$
- In the T1 system 24 voice channels are multiplexed in time therefore, data rate of a T1 stream **should be** = $24 \times 64\text{kbps} = 1.536\text{ Mbps}$

However, the **actual data rate** = 1.544Mbps

The extra 8000 bps ($1.544 - 1.536 = .008\text{Mbps}$) result from the overhead bits which are inserted alongside the data (details ahead).

T1 systems

4.93

- The T1 carrier system multiplexes binary codewords corresponding to samples of each of the 24 channels in a sequence. *A segment containing one codeword (corresponding to one sample) from each of the 24 channels is called a FRAME.*
- Each frame has $24 \times 8 = 192$ data bits and takes $125\mu\text{s}$.
- 12 consecutive frames is a **Superframe**
- At the receiver it is also necessary to know where a frame starts in order to separate information bits correctly. For this purpose, a **Framing bit** is added at the beginning of each frame. Therefore,

Total number of bits/ frame = 193

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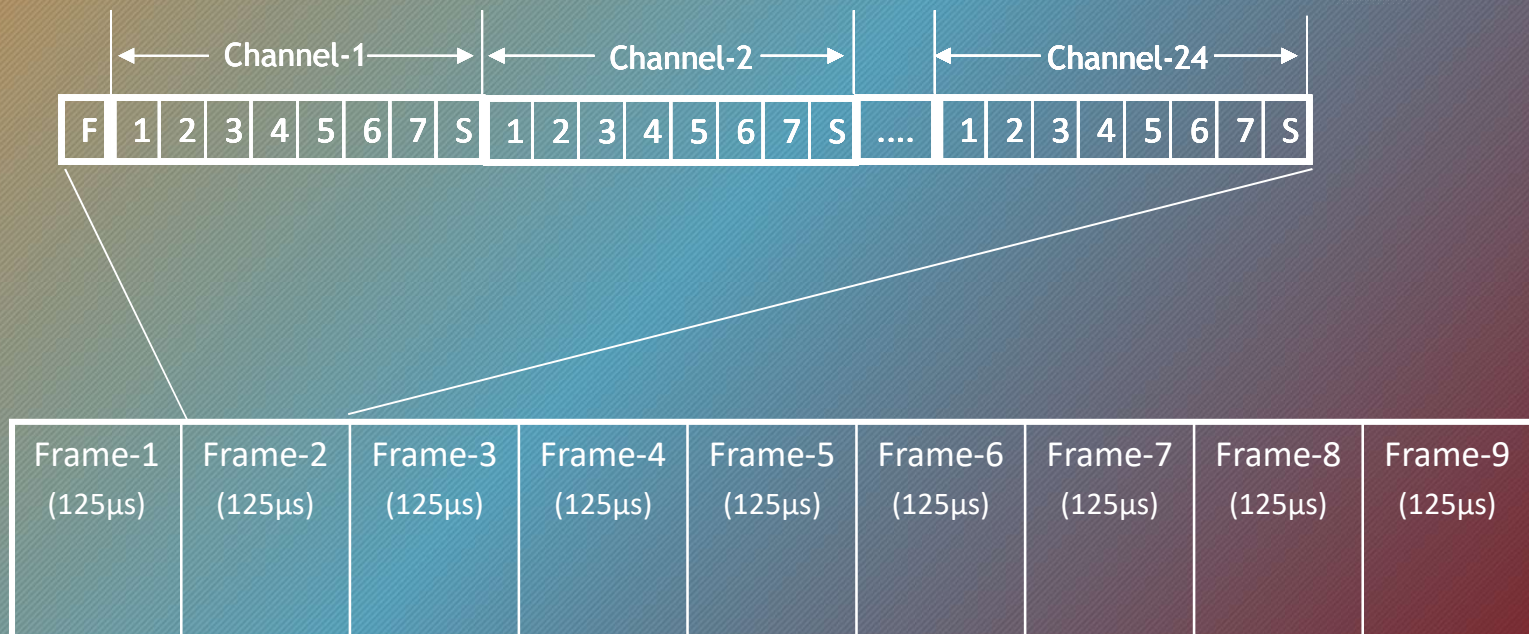
T1 systems

4.94

- Along with voice data, frames should also contain: Framing bits and Signaling bits.
- **Framing Bits:** Indicate start of frames.
- **Signaling Bits:** Contain control information such as Routing Information, On-Hook/ off-Hook signals, Alarm signals etc.
- Lets see how the T1 frame caters for these needs. We'll have a look at the frame structure of **older** T1 schemes first.

4.95

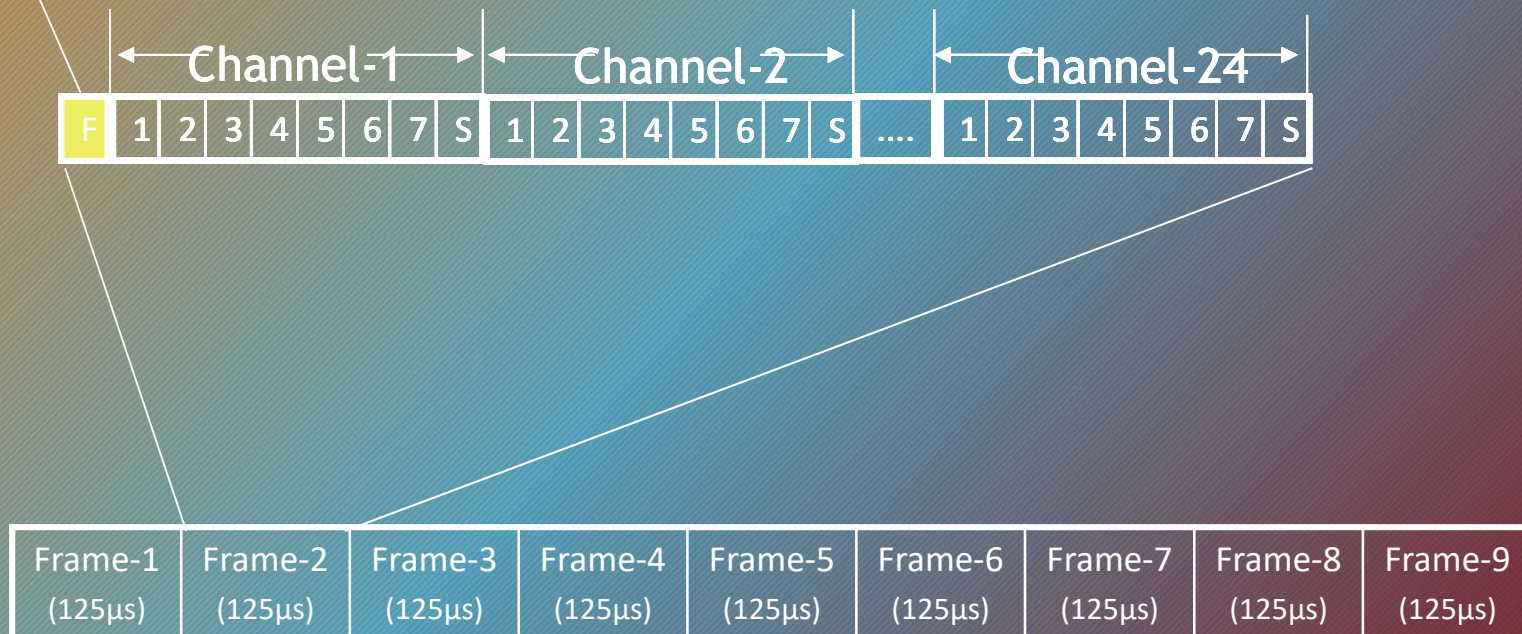
T1 frame format



T1 systems

4.96

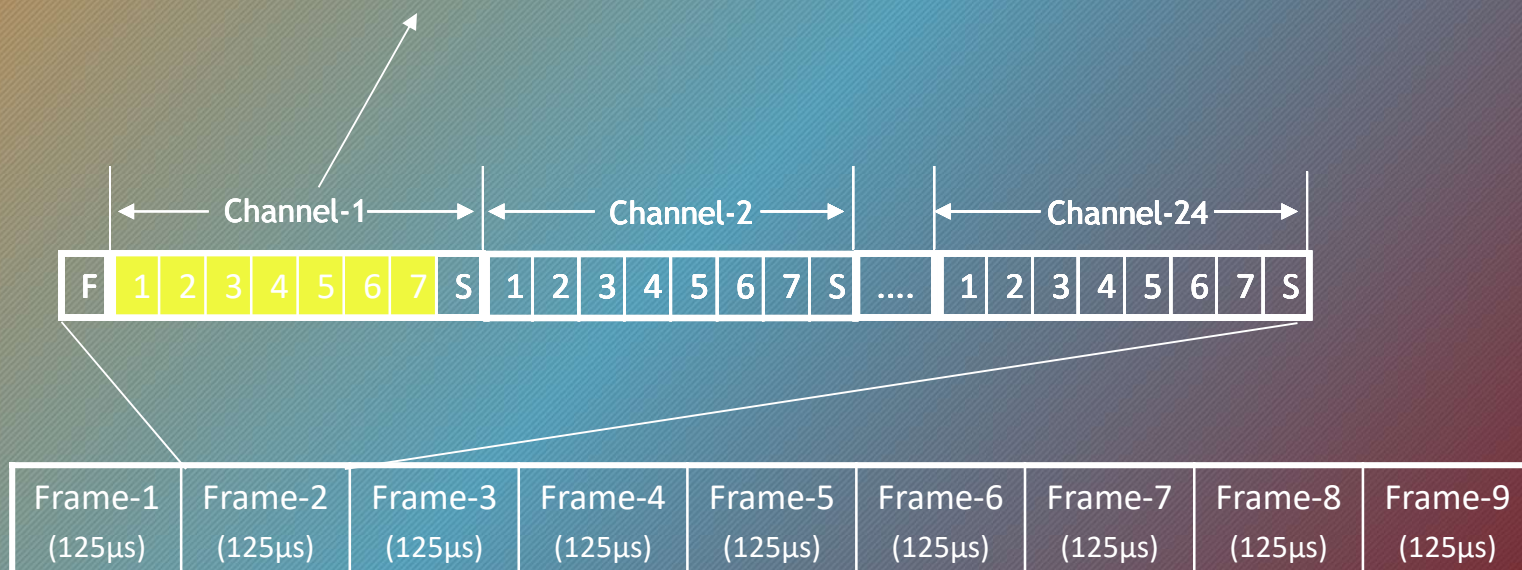
1 Framing Bit at
start of each frame



T1 Frame Format

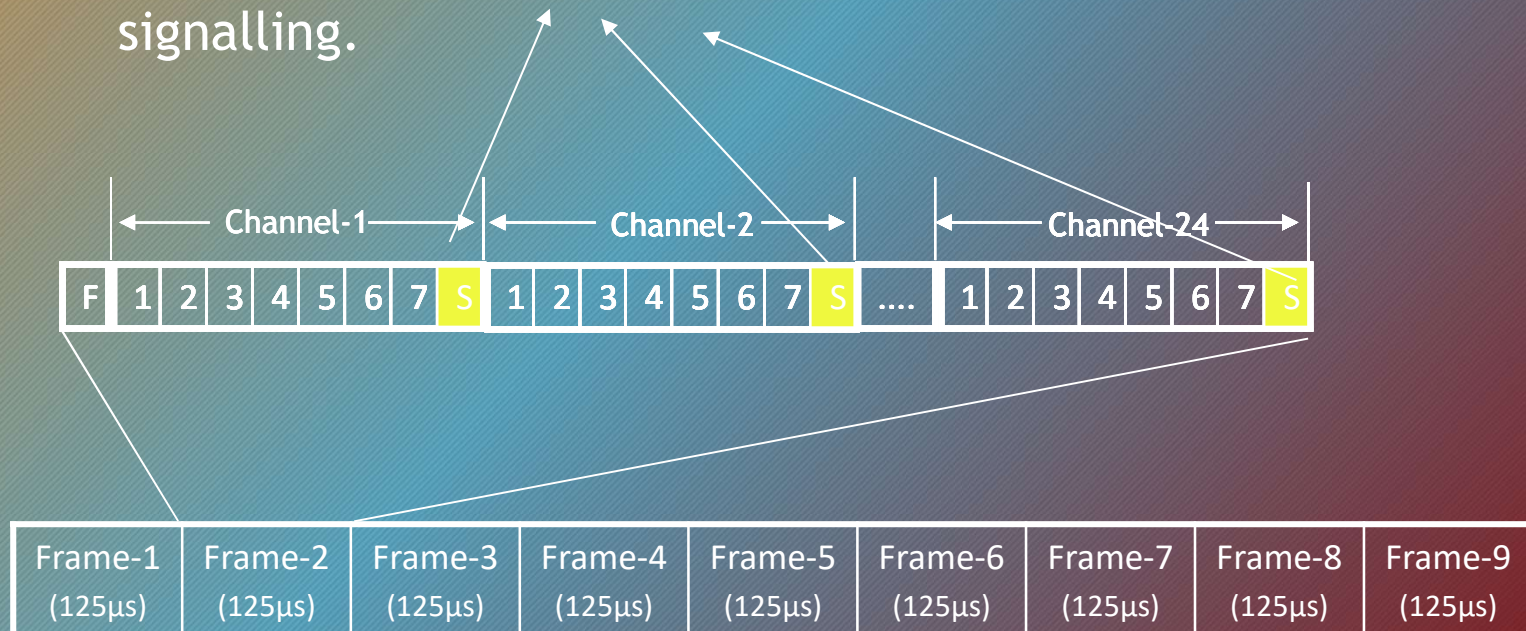
4.97

7 bits to represent each sample. Note this is the old T1 scheme and is now **obsolete**.



T1 Frame Format

1 bit in each channel reserved for signalling.
This means a total of **24 bits/ frame** for signalling.



T1 Frame Format

4.99

- **T1 newer version:**

24 signalling bits/ frame is just too much and that many bits are not required for voice data signalling.

This why a modification was introduced.

The newer version uses 8 bits to represent every sample. However, the total number of bits in each frame remains the same i.e, **193 bits/frame**.

- **Does this mean that the signalling bits are eliminated..???**
- **If so then how is the signalling information conveyed..????**

T1 Frame Format

4.10
0

- **T1 newer version:**

Actually the signaling bits are not eliminated altogether. Instead the number of signaling bits in a 'collection of frames' is reduced.

There are two T1 framing formats: **D4 superframe** and **extended super-frame (ESF)**

In the **D4 superframe**, signalling bits are included in every 6th frame and 12th frames

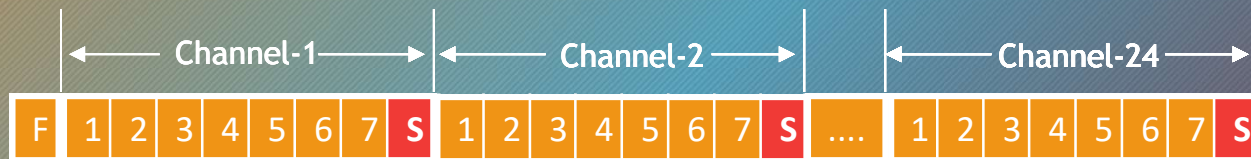
In the **ESF superframe**, signaling bits are included in every 6th, 12th, 18th and 24th frames.

This is done by replacing the Least Significant Bit (LSB) of every sample by a signaling bit. Therefore, some amount of information in every 6th, 12th, 18th, 24th sample is sacrificed for signaling. However, since only the LSB is removed there is no noticeable loss in voice quality.

4.10 1

T1 Frame Format

All other frames use 8 bits/sample. This technique in which signalling information is sent in every 6th and 12th frames is called **Robbed-Bit- Signalling**.



Frames 6, 12

Frame-1 (125μs)	Frame-2 (125μs)	Frame-3 (125μs)	Frame-4 (125μs)	Frame-5 (125μs)	Frame-6 (125μs)	Frame-7 (125μs)	Frame-8 (125μs)	Frame-9 (125μs)
--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------	--------------------



All Frames except 6, 12

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T1 Frame Format

4.10
2

- **Frame Synchronization in T1:**
 - It was mentioned in the previous slides that 1 Framing bit indicates the start of each frame. How is this framing bit different from any of the other bits that arrive before or after it?
 - One framing bit alone doesn't signify much. However, a collection of framing bits can be used to check the beginning of a collection of frames.
 - The receiver accumulates the framing bits of a collection of 12 frames and checks whether these 12 framing bits are the same as a pre-decided pattern (100011011100).
 - If the framing bits are exactly the same as this pattern then the transmitter and receiver are in frame Synchronization and the samples can be easily extracted from the frames.
 - What happens when the framing bits do not match the pre-decided pattern?

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D4 superframe (SF) signaling

4.10

Frame Number	Terminal Framing (TF)	Multi-frame Alignment Signal Fs	Signal Channel Destination
1	1	-	
2	-	0	
3	0	-	
4	-	0	
5	1	-	
6	-	1	A
7	0	-	
8	-	1	
9	1	-	
10	-	1	
11	0	-	
12	-	0	B

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ESF signaling

4.10

F bit functions FAS/DL/CRC

Multi-frame Number	Multi-frame Bit Number	FAS	DL	CRC	Signal Channel Designation
1	0	-	m	-	
2	193	-	-	e1	
3	386	-	m	-	
4	579	0	-	-	
5	772	-	m	-	
6	965	-	-	e2	A
7	1158	-	m	-	
8	1351	0	-	-	
9	1544	-	m	-	
10	1737	-	-	e3	
11	1930	-	m	-	
12	2123	1	-	-	B
13	2316	-	m	-	
14	2509	-	-	e4	
15	2702	-	m	-	
16	2895	0	-	-	
17	3088	-	m	-	
18	3281	-	-	e5	C
19	3474	-	m	-	
20	3667	1	-	-	
21	3860	-	m	-	
22	4053	-	-	e6	
23	4246	-	m	-	
24	4439	1	-	-	D

4.10 5

Direction	State	A	B	C	D
Transmit	On Hook	0	0	0	0
Transmit	Off Hook	1	1	1	1
Receive	On Hook	0	0	0	0
Receive	Off Hook	1	1	1	1

Figure 9: ABCD signaling bits in a T1 frame.

Khi đầu thu không nhận được tín hiệu tới → cảnh báo RAI (remote alarm indicator) → phát “11111111 00000000” trên DL (trong ESF) hoặc set các bit 2 trong mỗi ts xuống 0 (SF-D4).

T1 Frame Format

- **Loss of Frame Synchronization in T1:**

- When frame synchronization is lost in T1, the receiver immediately send an LFA alarm, indicating loss of frame synchronization, to the transmitter.
- Without disconnecting the calls the transmitter redirects the data and sends it via an alternate route, and also tries to regain frame synchronization with the receiving party by send some fixed pattern.
- Since it takes 12 frames to realize that frame synchronization is lost, it means that at-least 12 voice samples from all the effected 24 channels will be lost. However, this is not noticed by the subscribers because this is still a very small time frame

$12 \times 125\mu s = 1.5ms$ of conversation is lost.

Comparison between E1 and T1

Common characteristics	PCM-30 and PCM-24
Sampling frequency	8kHz
Number of samples per telephone signal	8000
Length of PCM frame	125 μ s
Number of bits in each word	8
Telephone channel bit rate	64kbit/s

Differing characteristics	PCM 30	PCM 24
Encoding / decoding	A-law	μ -law
Number of segments in characteristic	13	15
Number of time slots per PCM frame	32	24
Number of bits per PCM frame	256	193
Length of an 8-bit time slot	approx. 3.9 μ s	approx. 5.2 μ s
Bit rate of time-division multiplexed signal	2048kbit/s	1544kbit/s

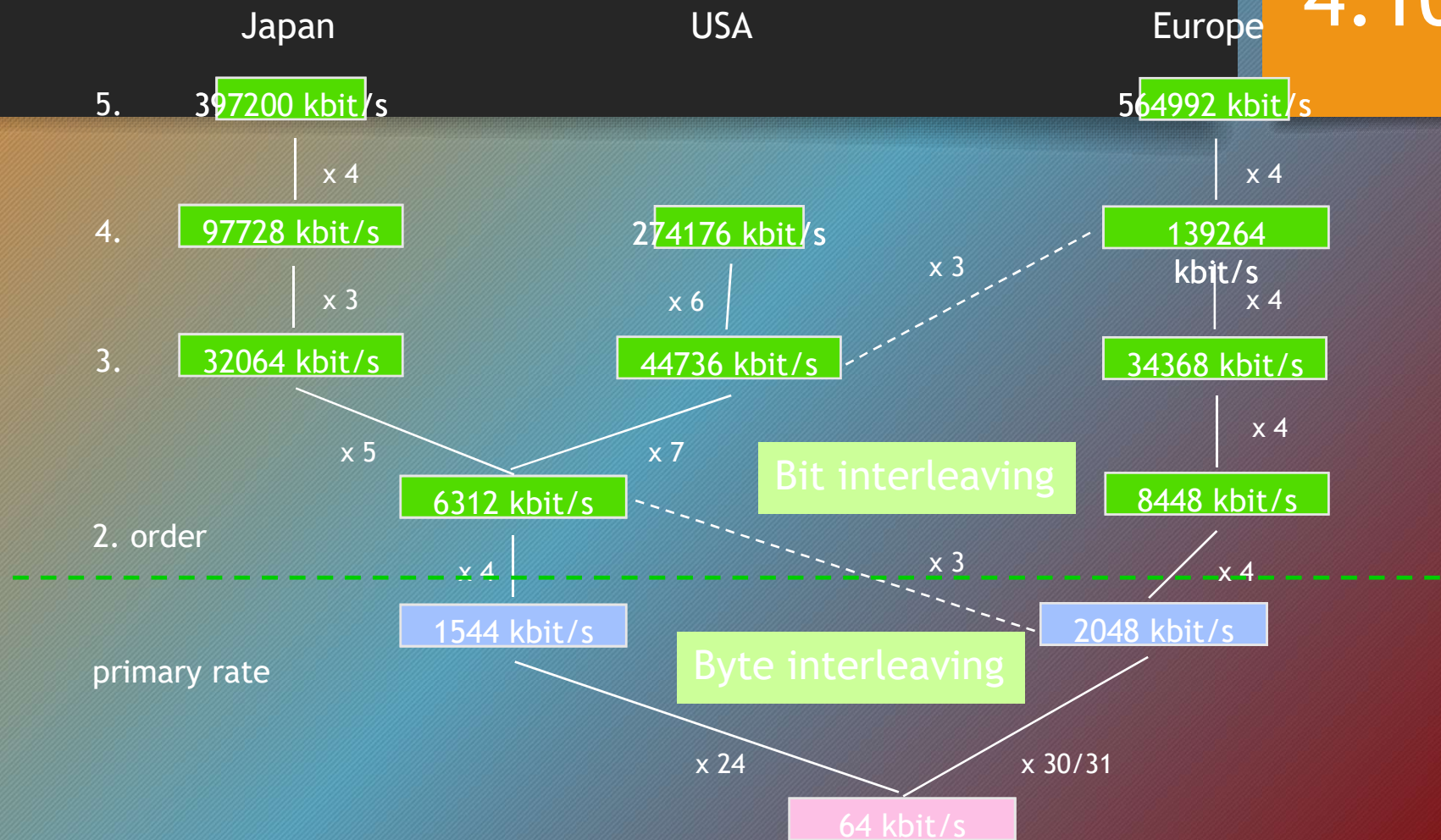
Advance digital transmission systems

4.108

- Synchronous Optical Networking (SONET) and Synchronous Digital Hierarchy (SDH) are standardized protocols that transfer multiple digital bit streams over optical fiber using lasers or highly coherent light from light-emitting diodes (LEDs).
- At low transmission rates data can also be transferred via an electrical interface. The method was developed to replace the Plesiochronous Digital Hierarchy (PDH) system for transporting large amounts of telephone calls and data traffic over the same fiber without synchronization problems.

PDH Systems Worldwide

4.109



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Question

4.110

1. What is the major overriding advantage of binary digital transmission? Give at least two secondary advantages.
2. If I'm transmitting 8000 frames per second, what is the duration of one frame?
3. Our system uses linear quantization. How can we reduce quantization distortion (noise)?
4. Present the diagram of time division multiplexing for E1.
5. Present digital transmission modes.
6. What is the function of times lost 0 and 16 of frame E1s?
7. How is the signaling in the data transmission standard T1?

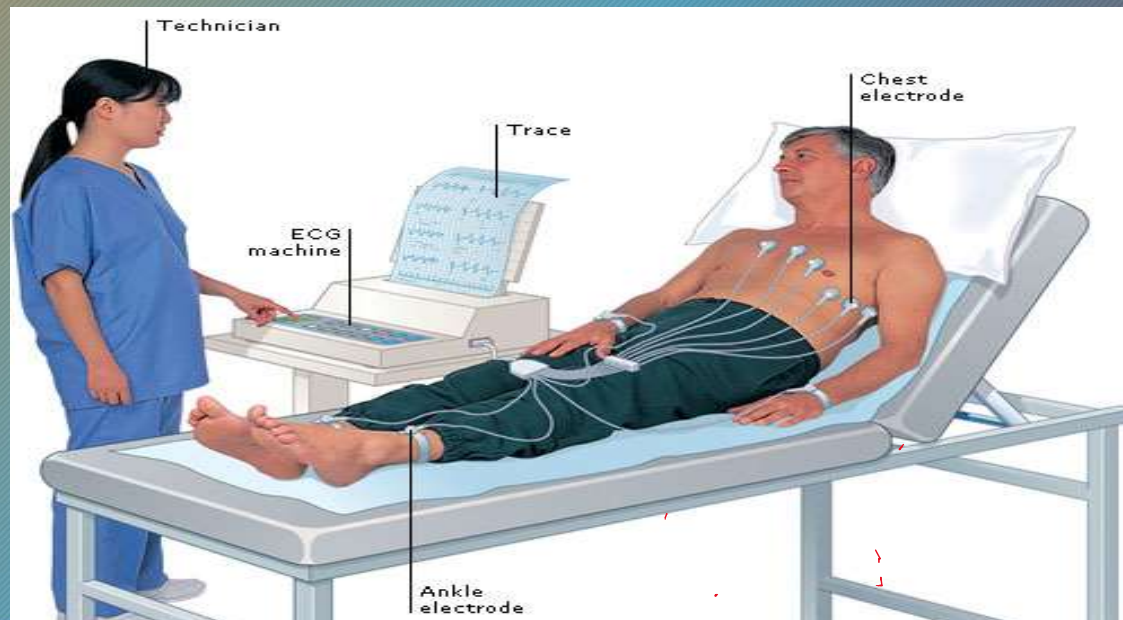
U_{ECG}

Exercises

4.111

1>Design a 24-lead EEG data acquisition system (24 EEG channels). Knowing that the bandwidth of the ECG signal is [0-100Hz], the receiver sensor has an output amplitude of [0-2mV]. Receiver system uses an 8-bit IC PCM for multiplexing. This IC allows you to multiplex 24 EEG channels with input voltage range [0-5V]. Perform the following requirements:

- Define the input/output of the design problem.
- Design block diagrams and define input and output parameters of blocks.



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Exercises

4.112

2. Design an audio studio recorder using 5.1 standard. It is known that the studio recorder use PCM multiplexer for 5 input audio signals with 8 bit/sample. The audio signal has bandwidth $[0 \dots 15\text{kHz}]$.

